SpecBraM: Efficient EEG Foundation Model Pretraining by Predicting Masked Band Power

Anonymous CVPR submission

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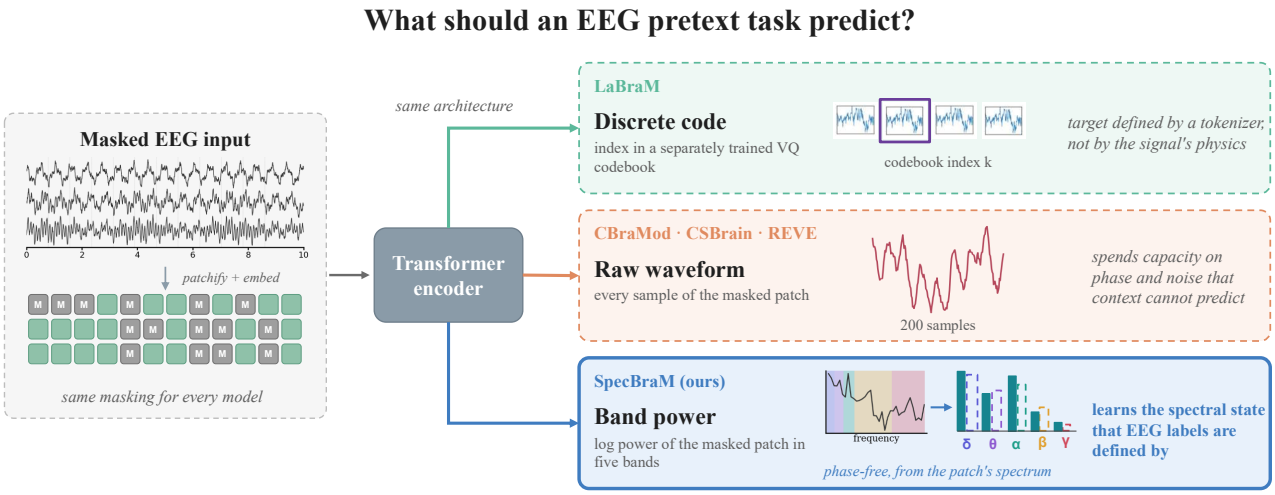


Figure 1. **What should an EEG foundation model predict?** The model families encode the same masked token grid with a transformer and differ in the pretraining target. LaBraM predicts the discrete codes of a separately trained vector-quantized tokenizer; CBraMod, CSBrain and REVE reconstruct every sample of the masked patch. SpecBraM predicts the log power of each masked patch in the five classical frequency bands, computed from the patch's spectrum: a phase-invariant physical quantity associated with the spectral state.

Abstract

EEG foundation models aim to decode brain states across clinical and cognitive tasks from few labels. They are pretrained on large unlabeled corpora with objectives borrowed from images and language, reconstructing the raw waveform of masked patches or predicting discrete tokens, and their progress has come mainly from more data and larger models. These objectives also fit waveform details beyond the spectral information relevant to many EEG labels. Sleep stages, clinical slowing and many cognitive states are associated with the power of neural rhythms, while a short patch also contains phase and sample-level variation that may be only partly predictable from context. We present ***SpecBraM****, an EEG foundation model pretrained by* masked band-power prediction*: from the visible context, it predicts the log power of each masked (channel, time) patch in the classical frequency bands, a parameter-free physical target computed from the patch's spectrum. Under an idealized spectral model, band-averaged power is sufficient for the band levels, and its logarithm stabilizes sampling variance when the numerical floor is ignored. This motivates a compact target for efficient learning. With backbone, data and training steps fixed, the trained models explain substantially more of the band-power target's variance than of the waveform's, and on sleep staging band-power pretraining reaches the linear-probe quality of waveform reconstruction with a fraction of the steps and with a*

model a tenth of the size. Under an identical strict linear probe on eight datasets spanning six task types, SpecBraM, pretrained on a fraction of the data of recent models, outperforms CBraMod, LaBraM and CSBrain on nearly all tasks and has higher point estimates than REVE, which uses far more data, on most tasks. On sleep staging it also leads the compared models with few labelled training epochs and in unsupervised clustering agreement.

1. Introduction

EEG is central to clinical neuroscience—sleep staging, screening for abnormal background activity, studies of depression and cognitive load—but expert labels are scarce, which motivates foundation models pretrained once on large unlabeled corpora such as the TUH EEG Corpus (TUEG) [25] and transferred to many tasks [11, 16, 33, 39]. Their pretraining objectives are borrowed from other modalities. LaBraM [16] trains a vector-quantized tokenizer and predicts its discrete codes, as BEiT does for images [5]; CBraMod [33], CSBrain [39] and REVE [11] reconstruct the raw waveform of masked patches, as masked autoencoders reconstruct pixels [14]. Progress has come mainly from larger models and more data: REVE pretrains on about 60,000 hours.

We present **SpecBraM**, an EEG foundation model whose pretraining target is a physical quantity rather than the signal. Spectral power informs many EEG labels: AASM sleep stages are specified by the amount of delta activity, spindles and alpha [6], clinical reports describe "background slowing", and quantitative EEG uses band powers and their ratios. SpecBraM therefore keeps the standard masked-patch transformer but, for each masked (channel, time) patch, predicts its log power in the five classical frequency bands, computed from the patch's own spectrum (Fig. 1). The target needs no tokenizer and no target network and costs one FFT per patch.

This target makes pretraining efficient, and we motivate a possible mechanism (Sec. 4). Under squared error, variation remaining after conditioning on the visible context contributes gradient variance but no expected gradient. In an idealized stationary-Gaussian Whittle model, marginal Fourier phases are ancillary for spectral parameters; this does not imply that context cannot predict phase. With exactly flat bands and observation restricted to those bands, the band-averaged periodogram is sufficient for the band levels and is their maximum-likelihood estimate. Its zero-floor logarithm has state-independent conditional sampling variance. These properties motivate a compact spectral target, but do not prove lower gradient noise or faster optimization for the implemented model.

The experiments establish the efficiency gains. With backbone, data and steps fixed, band power reaches after one to two pretraining epochs the linear-probe quality that waveform reconstruction reaches after sixteen, and a band-power model with 4.2M parameters surpasses a waveform model with 51.3M. Pretrained on 2,388 hours, SpecBraM is evaluated with an identical strict linear probe on eight datasets from six task types. Our contributions are:

- **SpecBraM**, an EEG foundation model pretrained by masked band-power prediction, a parameter-free physical target that replaces waveform reconstruction (Sec. 3).
- **An idealized rationale for the target**: band power summarizes state-associated spectral levels without requiring phase prediction; risk and gradient decompositions clarify the role of contextual uncertainty and target noise (Sec. 4, Appendix A).
- **Evidence across tasks and in efficiency.** Under a strict linear probe SpecBraM has higher point estimates than CBraMod, LaBraM and CSBrain on (nearly) all tasks and REVE on most of them, using less pretraining data than several baselines. Matched target studies show gains in pretraining steps, model size and low-label sleep probing (Sec. 6).

We do not claim a new paradigm: masked prediction of hand-crafted features is established in vision [34] and speech [15], and spectral information enters the targets of several EEG models (Sec. 2). FAME [38] and MANAS-2 [21], which also use band log-power targets, are concurrent work.

2. Related Work

EEG foundation models. Early self-supervised EEG models used contrastive or autoregressive objectives (BENDR [20], Neuro-GPT [10]). Most recent models pretrain a transformer on TUEG with a masked-prediction objective: CBraMod [33], CSBrain [39] and REVE [11] reconstruct the raw waveform of masked patches; LaBraM [16] predicts the codes of a vector-quantized tokenizer; EEGPT [32] combines reconstruction with latent alignment; NeuroLM [17] maps EEG to language tokens; BIOT [36] handles heterogeneous biosignals. These models differ mainly in architecture, scale and data, while their targets are inherited from vision and language. A waveform target asks the model to predict sample-level details, including phase, in addition to spectral structure; some of this variation may remain uncertain given the context. A learned codebook target is defined by a tokenizer rather than by a physiological quantity. Our target directly measures band power, a spectral quantity relevant to many EEG labels.

Spectral information in EEG pretraining. Spectral features enter several models as inputs or tokenizers: BIOT tokenizes STFT magnitudes, CBraMod adds a frequency-domain patch embedding, LaBraM's tokenizer reconstructs the Fourier amplitude and phase, BioSerenity-E1 [7] and BandVQ [30] quantize log spectra, and BrainBERT [31] re-

constructs intracranial spectrograms. Concurrently, FAME [38] decodes band-averaged log power of the whole sequence to counter the low-frequency bias of masked autoencoders, and MANAS-2 [21] adds band log-power to waveform reconstruction. In these works spectral information is an input, a quantized code or an auxiliary term, and its role as the target is neither motivated nor isolated. We use band power of the masked patches as the sole target, derive it from a likelihood argument, and isolate its effect with matched controls.

Feature and latent prediction. In vision and speech, predicting fixed hand-crafted features of masked inputs (HOG in MaskFeat [34], MFCC clusters in HuBERT [15]) can outperform predicting the raw signal, and latent-predictive methods (I-JEPA [2], data2vec [3], EEG2Rep [12]) predict learned representations but need anti-collapse mechanisms such as SIGReg [4]. Band power plays the role of such a feature for EEG: it is a physiological quantity relevant to many labels, and we use the fixed target without an explicit anti-collapse regularizer.

3. SpecBraM

Input and tokens. A 10-s window at 256 Hz over the 19 channels of the 10–20 montage, normalized per recording (median and interquartile range), is cut per channel into non-overlapping patches of $P{=}200$ samples (0.78 s), giving a token grid with $C{=}19$ channels and $T_p{=}\lfloor 2560/200 \rfloor{=}12$ time patches; the last 160 samples (0.625 s) of each window are not used, during pretraining and in every downstream window (including each of the three 10-s windows of a 30-s sleep epoch; 5-s TUEV windows give six patches). The main model embeds each patch with a single linear map (as CBraMod does). A variant decomposes each patch with a learnable 5-band filterbank before embedding ("filterbank tokenizer"); we report both because the target study crosses tokenizer and target.

Encoder and masking. A 12-layer criss-cross transformer ($D{=}512$, 8 heads; 51.3M parameters) alternates attention along the channel and time axes, with learned channel and time position embeddings. Each (channel, time) patch is masked independently with probability 0.3; the embeddings of masked patches are replaced by a learned mask token.

Target. For each patch $\boldsymbol{x}_{c,t} \in \mathbb{R}^P$ of the uncorrupted signal, let $X_k = \sum_\tau x_{c,t,\tau} e^{-i2\pi k\tau/P}$ be its discrete Fourier transform (rectangular window, no padding) at $f_k = kf_s/P$. For the five classical bands $B_1, \ldots, B_5 = [0.5, 4), [4, 8), [8, 13), [13, 30), [30, 45)$ Hz ($\delta, \theta, \alpha, \beta, \gamma$),

$$P_{c,t,b} = \frac{2}{P^2} \sum_{f_k \in B_b} |X_k|^2,$$
$$y_{c,t,b} = \log\big(P_{c,t,b} + \varepsilon\big), \qquad b = 1, \ldots, 5, \qquad (1)$$

with $\varepsilon = 10^{-3}$ in the robust-scaled input units; its effect depends on the band power relative to this floor. By one-sided Parseval, $P_{c,t,b}$ is the variance of the patch within band b; it equals the band-averaged periodogram of Sec. 4 times the per-band constant $2m_b/P$, where m_b is the number of Fourier bins in the band (3/3/4/13/12 at $\Delta f = 1.28$ Hz; Table 7). The target is fixed (no learnable parameters) and patch-local: it uses no samples of neighbouring, possibly visible patches.

Choice of bands. The classical partition reflects physiologically distinct rhythms (slow-wave activity, theta, the posterior alpha rhythm and sleep spindles at its upper edge, beta and gamma) and is the vocabulary in which sleep scoring and clinical EEG reports are written. Its exact edges are conventions (alpha is given as 8–12 or 8–13 Hz, and individual alpha frequency varies by several hertz), so we treat them as a sensible default rather than an optimum.

Objective. A linear head maps each encoder output $H_{c,t}$ to five predictions $\hat{y}_{c,t,b}$. With $\mathcal{M}$ the set of masked patches,

$$\mathcal{L} = \frac{1}{5|\mathcal{M}|} \sum_{(c,t)\in\mathcal{M}} \sum_{b=1}^{5} \big(\hat{y}_{c,t,b} - y_{c,t,b}\big)^2, \qquad (2)$$

the mean squared error over the masked patches and the five bands. There is no reconstruction term, no latent-prediction term and no regularizer. The target is fixed; this alone does not guarantee a noncollapsed representation. The waveform-reconstruction controls replace the loss of Eq. (2) by the MSE between a linear decoder output and the raw samples of each masked patch ("raw") or five band-pass-filtered versions of the patch waveform (fixed 65-tap Hann-windowed cosine filters centred at 2/6/10/21/38 Hz; "band-waveform"), with everything else unchanged.

4. Why Predicting Band Power Is Efficient

We motivate the target under an idealized model; the statements and proofs are in Appendix A. Before the logarithm and numerical floor, the main target, Eq. (1), is a per-band constant times the band-averaged periodogram. The exact log-noise identities below apply to the zero-floor idealization with independent Fourier bins and exactly flat in-band spectra. The implemented $\log(P + \varepsilon)$ and real EEG need not satisfy them. The analysis motivates the target and suggests empirical hypotheses; it is not a proof of optimality or an optimization-speed guarantee.

Model. Given a brain state θ, a patch is a segment of a zero-mean stationary Gaussian process with spectral density S_θ; the state acts on the data only through S_θ. Asymptotically, the discrete Fourier coefficients at distinct frequencies are independent circular complex Gaussians, so the periodogram ordinates are $I_k \approx S_\theta(f_k)E_k$ with $E_k \sim \mathrm{Exp}(1)$, and the phases are uniform and independent of θ [8]. The Whittle likelihood [35] depends on the data only through $\{I_k\}$.

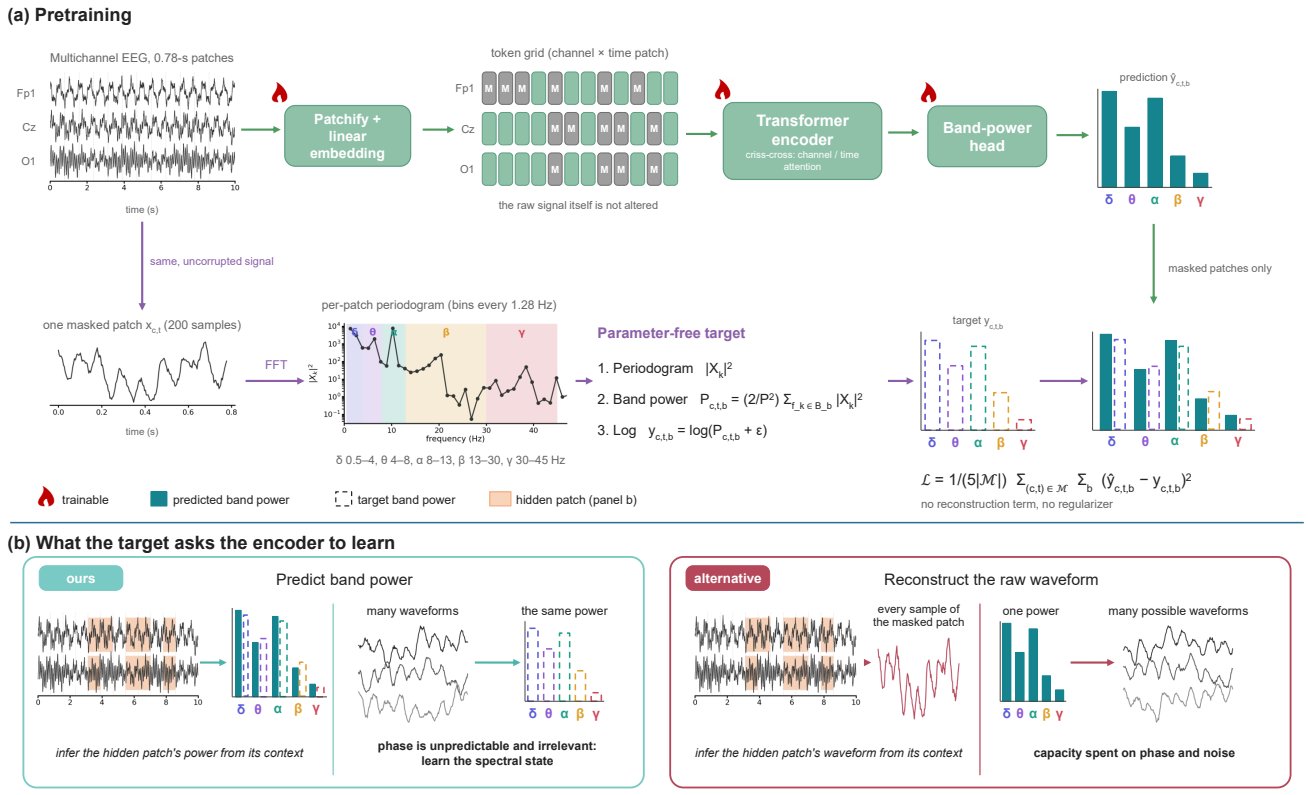


Figure 2. **SpecBraM pretraining.** (a) Each channel is cut into 0.78-s patches that are embedded linearly; 30% of the (channel, time) tokens are replaced by a learnable [M] token, while the signal itself is not altered. A criss-cross transformer encodes the token grid and a linear head predicts five log band powers for every masked token. The target is computed from the periodogram of the uncorrupted patch and has no parameters. (b) Predicting band power asks the encoder to infer the spectral state of a hidden patch from its context; reconstructing the waveform also asks for sample-level details, including phase, that the context may only partly determine.

What band power keeps. (i) In the Whittle model, the periodogram is sufficient for its spectral parameters and the marginal phases are ancillary (Prop. 1); this does not establish independence from visible context. (ii) With exactly constant in-band spectra and observation restricted to those bins, the band-averaged periodogram is sufficient for the band levels and is their maximum-likelihood estimate (Prop. 2). (iii) For its zero-floor logarithm, $\mathrm{Var}[\log P_b \mid \theta] = \psi'(m_b) \approx 1/m_b$ is independent of the state (Prop. 3). This motivates variance stabilization, but neither sufficiency for arbitrary EEG labels nor an exact noise model for Eq. (1) follows.

A possible learning advantage. Under squared error, prediction risk splits into error relative to $g(\boldsymbol{c}) = \mathbb{E}[Y \mid \boldsymbol{c}]$ and the Bayes risk given the fixed context (Prop. 4). Only the former can be reduced by improving the predictor. Under an additional independent-patch idealization, the zero-floor band-power predictor is $\mathbb{E}[\log s_b(\theta) \mid \boldsymbol{c}] + \text{const}$, a posterior average of spectral levels, rather than the state itself. Its conditional variance contains both $\psi'(m_b)$ and posterior uncertainty about $\log s_b(\theta)$. In this same independence model the zero-mean waveform predictor would be zero. A separate correlated Gaussian model instead yields Wiener interpolation conditional on the state, with additional uncertainty when the state is unknown. For either target, the full conditional covariance contributes Jacobian-weighted gradient noise (Prop. 5). Standard SGD upper bounds can worsen with gradient variance [13], but the two targets also differ in conditional mean, dimension, scale and Jacobian. These identities suggest a possible advantage of a compact spectral target; they do not establish an ordering of gradient noise or steps to a given downstream score.

Predictions. (P1) At matched backbone, data and steps, band power should give representations that are easier to read out linearly, especially with few labels. (P2) It should reach a given quality with fewer pretraining steps and smaller models. (P3) Reconstructing band-filtered *waveforms* need not yield the same gains, since they still require sample-level prediction. (P4) The advantage should shrink for tasks that depend on transient waveform shape or cross-channel phase relations. Sec. 6 tests these.

5. Experimental Setup

Pretraining. TUEG v2.0.1 at 256 Hz, 10-s windows, per-recording robust normalization, with recordings shorter than 5 min dropped, 60 s trimmed from each end and windows exceeding 150 μV rejected. All models use a random subset of 2,100 patients (859,606 windows, 2,388 h; the same subset for every run), AdamW (learning rate 1.7×10^{-3}, weight decay 0.05), 7 warm-up epochs and a cosine schedule defined over 105 epochs, stopped after epoch 16 for every run (matched steps), effective batch 512 on four RTX A6000 GPUs. Pretraining seeds 42/43/44 change initialization and data order on the same subset. Downstream patients are not excluded from pretraining (as in prior work).

Strict linear probe (primary metric). The encoder is frozen; the tokens of a window are mean-pooled; features are standardized on the training split; a class-balanced logistic regression is fitted with $C \in \{0.01, 0.1, 1\}$ selected on the validation split by balanced accuracy, and the test split is scored once. For sleep, a 30-s epoch is the mean of its three 10-s window features. The probe is deterministic, so baselines with a single released checkpoint need no repetitions; our models are reported over three pretraining seeds. Sleep label efficiency reuses the full-label-selected C on 1–50% of the training epochs (five fixed random subsets, identical for all models); these fractions do not include the labels used for hyperparameter selection and do not subsample subjects. Each model receives the same samples in its own documented input convention: ours at 256 Hz with per-recording robust scaling on the 19 standard leads; CBraMod and LaBraM at 200 Hz, raw μV/100, with the recording's own channels (LaBraM by its channel names); BIOT with its 95th-percentile scaling on C3/C4 fed to its C3–A2/C4–A1 channel tokens; CSBrain as CBraMod, with the brain-region layout of its own wrapper for each dataset; REVE at 200 Hz with per-recording z-scoring clipped at ±15 SD and its own electrode-position bank. We compare with released models of comparable size (3–70M parameters: BIOT, CBraMod, LaBraM-Base, CSBrain, REVE-Base; ours 51M); REVE-Large (390M) belongs to a different scale class and is not included. Every baseline is measured with this probe; no linear-probe number is quoted from other papers. For REVE, the readout (mean pooling, or its own pretrained attention pooling; both frozen and fitted without downstream labels) is selected using validation labels, like C.

Downstream tasks. We evaluate on eight public datasets from six task types: sleep staging (ISRUC [19], HMC [1]; five stages), clinical EEG (abnormality detection on TUAB and six-class event classification on TUEV [25]), mental workload (MA [40]), depression (Mumtaz [24]), emotion recognition (FACED [9], nine classes) and motor imagery (PhysioNet-MI [27], four classes). We reproduce the preprocessing and split protocols of CBraMod and CSBrain, with matching sample counts. In TUAB's original per-class training/validation split, three patients occur in both splits; its test patients are disjoint from both. This limits validation independence and is separate from possible TUEG pretraining overlap. TUEV has no patient overlap among its downstream splits. Appendix C describes every dataset.

6. Results

6.1. Strict linear probe across tasks

Table 1 summarizes the strict linear probe on all tasks, grouped by task type. As for the external models, which are released as single checkpoints, we evaluate one checkpoint of SpecBraM, our main released model, and give every model a 95% confidence interval from 1,000 bootstrap resamples of the test subjects (TUEV: recordings). SpecBraM has higher point estimates than CBraMod and LaBraM on every task, BIOT on sleep staging, and CSBrain on every task except event classification (TUEV 0.555 vs. 0.564). REVE, of similar size but pretrained on about 25 times more data, is the strongest baseline. On *sleep staging* SpecBraM has no detected difference on ISRUC (0.790 vs. 0.793) and a higher point estimate on HMC (0.742 vs. 0.704); on *workload*, *depression*, *emotion* and *motor imagery* its point estimates are higher (mental arithmetic 0.781 vs. 0.684, Mumtaz 0.939 vs. 0.902, FACED 0.221 vs. 0.200, PhysioNet-MI 0.417 vs. 0.403); on *clinical EEG* there is no detected difference on event classification (TUEV 0.555 vs. 0.546), and its abnormality-detection point estimate is lower (TUAB 0.813 vs. 0.821). The intervals are wide for the small cohorts (mental arithmetic has four test subjects, Mumtaz eleven), and the TUH tasks share corpus provenance with TUEG; their patients were not excluded from our pretraining. Table 8 reports nominal, unadjusted paired bootstrap intervals for the differences (Appendix C). Intervals containing zero indicate no detected difference under this procedure, not equivalence. Appendix C reports Cohen's κ, weighted F1 and, for binary tasks, AUROC and AUC-PR for every dataset, together with the mean and standard deviation over our three pretraining seeds.

Model size, input convention and readout also differ between the models; varying each in turn (Appendix G) does not close the gap to CBraMod and LaBraM. Although our focus is frozen use, SpecBraM also fine-tunes well: with full fine-tuning on sleep staging it reaches 0.818/0.770 balanced accuracy on ISRUC/HMC, above the listed published point estimates of CBraMod, CSBrain and REVE under their respective recipes (Appendix E).

6.2. Efficiency

Pretraining steps. Table 2 follows the linear probe along pretraining for the two targets, with everything else fixed.

Table 1. **Strict linear probe across task types** (balanced accuracy). Frozen encoder, mean-pooled tokens, logistic regression with C selected on the validation split, test scored once; every model sees the identical samples and splits in its own input convention (Sec. 5), and every number is measured by us with the publicly released weights. Every model is one checkpoint: SpecBraM is our main released checkpoint, the baselines are their released ones. Brackets: 95% confidence interval from 1,000 bootstrap resamples of the test subjects (TUEV: recordings); the spread over our three pretraining seeds is given in Appendix C. REVE: readout selected on the validation split. BIOT's released channel tokens cover only the sleep montage. Per-dataset results with Cohen's κ, weighted F1 and AUROC/AUC-PR: Appendix C. Bold: best per column.

		Sleep staging		Clinical EEG		Workload	Depression	Emotion	Motor imagery
Model	Pretraining	ISRUC	HMC	TUAB	TUEV	MA	Mumtaz	FACED	PhysioNet-MI
SpecBraM	2,388 h	0.790 [0.764, 0.816]	**0.742** [0.699, 0.776]	0.813 [0.775, 0.850]	0.555 [0.491, 0.648]	**0.781** [0.722, 0.840]	**0.939** [0.868, 0.995]	**0.221** [0.195, 0.248]	**0.417** [0.384, 0.450]
CBraMod	∼9k h	0.728 [0.702, 0.752]	0.619 [0.576, 0.654]	0.778 [0.738, 0.814]	0.525 [0.471, 0.609]	0.653 [0.507, 0.799]	0.893 [0.760, 0.988]	0.171 [0.156, 0.188]	0.312 [0.286, 0.340]
LaBraM-Base	∼2.5k h	0.752 [0.720, 0.777]	0.632 [0.583, 0.673]	0.776 [0.740, 0.812]	0.468 [0.416, 0.545]	0.681 [0.597, 0.736]	0.876 [0.744, 0.986]	0.167 [0.152, 0.181]	0.299 [0.285, 0.314]
BIOT	mixed	0.736 [0.691, 0.771]	0.658 [0.609, 0.697]	–	–	–	–	–	–
CSBrain	∼9k h	0.730 [0.706, 0.752]	0.611 [0.580, 0.641]	0.772 [0.736, 0.805]	**0.564** [0.505, 0.621]	0.556 [0.403, 0.653]	0.864 [0.774, 0.943]	0.179 [0.161, 0.199]	0.298 [0.273, 0.325]
REVE-Base	∼60k h	**0.793** [0.765, 0.816]	0.704 [0.664, 0.737]	**0.821** [0.781, 0.861]	0.546 [0.500, 0.587]	0.684 [0.562, 0.771]	0.902 [0.743, 0.994]	0.200 [0.179, 0.225]	0.403 [0.371, 0.438]

Table 2. **Along pretraining** (linear patch embedding, seed 42, strict linear probe BA; bold: better target at that epoch).

	ISRUC		HMC	
Epoch	BP	Raw	BP	Raw
1	**0.7716**	0.7398	**0.7125**	0.6593
2	**0.7812**	0.7411	**0.7222**	0.6664
4	**0.7866**	0.7431	**0.7276**	0.6689
8	**0.7904**	0.7572	**0.7477**	0.7140
12	**0.7904**	0.7688	**0.7364**	0.7160
16	**0.7901**	0.7627	**0.7420**	0.7152

Table 3. **How much of its pretraining target each model predicts**: fraction of the target variance explained, $R^2 = 1 - \mathrm{MSE}/\mathrm{Var}$, on the masked patches of 4,000 held-out windows (TUAB evaluation split, not excluded from TUEG pretraining; masks as in pretraining), along pretraining (seed 42) and at the end over three seeds. BP: mean of the five per-band R^2 values; waveform: pooled-sample R^2. These are achieved scores, not estimates of Bayes ceilings. Band power, per band $\delta/\theta/\alpha/\beta/\gamma$ at epoch 16: 0.89 / 0.89 / 0.92 / 0.94 / 0.95.

Epoch	Band power	Raw waveform
1	0.801	0.012
2	0.880	0.016
4	0.902	0.021
8	0.909	0.025
12	0.916	0.028
16	0.917	0.031
16, three seeds	0.917 $\pm$0.001	0.030 $\pm$0.000

Table 4. **Label efficiency** (strict linear probe, balanced accuracy vs. fraction of labelled training epochs; <100%: mean of five fixed random subsets shared by all models). SpecBraM: main checkpoint. Bold: best per column.

Model	1%	5%	10%	25%	50%	100%
ISRUC						
BIOT	0.655	0.703	0.716	0.728	0.734	0.736
CBraMod	0.660	0.709	0.717	0.726	0.727	0.729
LaBraM-Base	0.680	0.724	0.730	0.741	0.746	0.752
CSBrain	0.633	0.693	0.708	0.721	0.726	0.730
REVE-Base	0.736	0.771	0.778	0.788	**0.791**	**0.792**
SpecBraM	**0.760**	**0.779**	**0.788**	**0.791**	**0.791**	0.790
HMC						
BIOT	0.582	0.625	0.638	0.648	0.655	0.658
CBraMod	0.539	0.590	0.602	0.613	0.616	0.620
LaBraM-Base	0.574	0.616	0.621	0.630	0.631	0.633
CSBrain	0.509	0.581	0.595	0.608	0.612	0.612
REVE-Base	0.635	0.679	0.691	0.700	0.702	0.704
SpecBraM	**0.662**	**0.698**	**0.717**	**0.733**	**0.738**	**0.742**

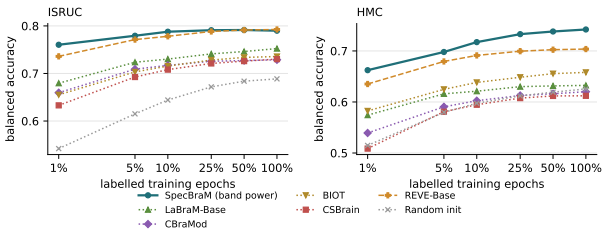


Figure 3. **Label efficiency** under the strict linear probe (balanced accuracy; SpecBraM: main checkpoint).

After a single epoch, band power reaches 0.772 on ISRUC, above raw-waveform reconstruction after sixteen epochs (0.763); on HMC it is close after one epoch (0.713 vs. 0.715) and exceeds it after two (0.722). Band power thus reaches the final quality of waveform reconstruction with 8 to 16 times fewer pretraining steps (P2). Target computation uses one FFT per patch, the head has five outputs per patch instead of 200, no tokenizer is trained beforehand (as for LaBraM) and no target network is maintained (as for latent prediction).

Table 5. **Clustering evaluation**: k-means (k=5) on frozen sleep features of the test split (standardized, 64 PCA dimensions; cluster fitting uses no labels), agreement with the expert stages. Mean over five k-means seeds; our rows: seed-42 checkpoints. REVE: readout selected by validation NMI using labels. Bold: best per column.

	ISRUC		HMC	
Model	NMI	ARI	NMI	ARI
SpecBraM	**0.417**	**0.310**	**0.236**	**0.179**
Raw-waveform recon. (same arch.)	0.401	0.278	0.182	0.135
CBraMod	0.277	0.184	0.083	0.056
LaBraM-Base	0.356	0.291	0.083	0.051
BIOT	0.149	0.091	0.057	0.025
CSBrain	0.263	0.166	0.069	0.038
REVE-Base	0.323	0.279	0.155	0.103

Achieved target prediction. Table 3 measures prediction on TUAB evaluation windows with the masks of pretraining. These windows are held out from downstream training; exclusion from pretraining is not established. The band-power model explains 80% of its target's variance after one epoch and 92% after sixteen; the waveform model explains 1.2% after one epoch and 3.1% after sixteen. Band-power R^2 is averaged over the five per-band scores, whereas waveform R^2 pools the sample coordinates. These are achieved scores, not Bayes predictability ceilings: residual error combines contextual uncertainty with model and optimization error, and does not determine a fraction of gradient noise. The early improvement of band-power prediction accompanies the early rise of its linear probe.

Model size and data. At 4.2M parameters, the size of CBraMod, the band-power model (0.788/0.728 on ISRUC/HMC) surpasses the 51.3M-parameter waveform model (0.770/0.715; Table 6), and it exceeds CBraMod, pretrained on about four times more data, by 6.0/10.9 points. SpecBraM is pretrained on 2,388 hours, about a quarter of the data of CBraMod and CSBrain and a twenty-fifth of REVE's, and ranks first on most tasks of Table 1.

Labels. On sleep staging, SpecBraM remains effective with few labelled training epochs (Table 4, Fig. 3). With 1% of the training epochs SpecBraM reaches 0.760/0.662 on ISRUC/HMC, more than BIOT, CBraMod, LaBraM and CSBrain obtain with all labels, and against 0.736/0.635 for REVE with 1%. It approaches REVE's full-training-label score with 25% of the training epochs on ISRUC (0.791 vs. 0.792) and exceeds it with 10% on HMC (0.717 vs. 0.704), and on HMC it leads every external model at every label fraction.

Unsupervised clustering. To assess the representation without a supervised classifier, we cluster the frozen sleep features with k-means and compare the clusters with the expert stages (Table 5). SpecBraM's features are organized by sleep stage more than those of any other model: NMI 0.417/0.236, against 0.401/0.182 for the same-architecture raw-reconstruction model and at most 0.356/0.155 for the external models. These results support sleep-stage structure in the frozen features. Clustering is unsupervised, but REVE's readout is selected by validation NMI using stage labels; this comparison is therefore not entirely label-free.

6.3. Controlled target study

Table 6 changes only the pretext target, with backbone, data, steps and launcher fixed, and reports the mean over three pretraining seeds. With the linear patch embedding the band-power target reaches 0.7965/0.7433 balanced accuracy on ISRUC/HMC, against 0.7701/0.7153 for raw-waveform reconstruction (P1). The difference is +2.6/+2.8 points with the linear embedding, +2.5/+1.7 with a learnable filterbank tokenizer and +3.5/+5.5 at 4.2M parameters; paired by pretraining seed, it is positive on all eighteen seed–dataset pairs, and it grows to +3.2 to +6.3 points with 1% of the labels. Reconstructing five band-pass-filtered waveforms instead of the raw one gives lower mean scores (0.7646/0.7173 vs. 0.7716/0.7201; P3). This is consistent with benefits from an aggregated, phase-invariant target, but does not isolate phase removal: output dimension and loss scale also change. Beyond sleep, band power has higher mean scores on mental arithmetic, depression screening and event classification, and close scores on the other tasks (Appendix D). Both studied spectral estimators give strong sleep results (Appendix F).

7. Limitations

The implemented target. With 0.78-s patches the frequency resolution is 1.28 Hz, so the δ band has only three bins (1.28/2.56/3.84 Hz) and activity below about 1 Hz, including much of slow-wave activity, is barely represented; the zero-floor idealization gives $\mathrm{sd}[\log P_b \mid \theta] \approx 0.63$ for δ and θ (Table 7). Longer patches, multitaper estimates (which give higher mean scores at 1% of the labels, Table 21) or per-band standardization (as in FAME [38]) are natural refinements. The classical band edges are conventions, and we did not search over partitions.

Probing protocol. Our probe mean-pools tokens and fits a linear classifier, which is strict and identical across models but is not each model's preferred readout; a per-channel readout raises CBraMod and LaBraM noticeably (Appendix G), and tasks that depend on spatially localized activity may favour other readouts.

Scope. The advantage of the target is clearest for sleep staging and is smaller or absent on emotion recognition, motor imagery and abnormality detection, and REVE, pretrained on 25 times more data, has a higher point estimate on abnormality detection. Several cohorts are small (MA has four

Table 6. **Pretext-target study** at matched backbone, data (2,388 h), steps (16 epochs) and launcher; only the target (or the tokenizer) changes. Strict linear probe, mean±std over three pretraining seeds unless stated. "1%": balanced accuracy with 1% of the labelled training epochs (mean of five fixed subsets). Δ: difference to the band-power row of the same block, in balanced-accuracy points; paired by pretraining seed, band power exceeds raw-waveform reconstruction on every seed in every block. Bold: best per column.

	BA (100% labels)		BA (1% labels)		Δ BA (pts)	
Target	ISRUC	HMC	ISRUC	HMC	ISRUC	HMC
Linear patch embedding, 51.3M parameters (main model)						
Band power (ours)	**0.7965** ±0.0059	**0.7433** ±0.0019	**0.765** ±0.006	**0.670** ±0.008	–	–
Raw-waveform reconstruction	0.7701 ±0.0076	0.7153 ±0.0028	0.721 ±0.017	0.617 ±0.017	-2.6	-2.8
Linear patch embedding, 4.2M parameters (CBraMod's size)						
Band power (ours)	0.7882 ±0.0053	0.7284 ±0.0040	0.741 ±0.021	**0.670** ±0.012	–	–
Raw-waveform reconstruction	0.7537 ±0.0042	0.6738 ±0.0083	0.709 ±0.008	0.607 ±0.002	-3.5	-5.5
Filterbank tokenizer						
Band power (ours)	0.7961 ±0.0067	0.7371 ±0.0059	0.764 ±0.003	0.658 ±0.016	–	–
Raw-waveform reconstruction	0.7716 ±0.0058	0.7201 ±0.0087	0.707 ±0.026	0.623 ±0.008	-2.5	-1.7
Band-waveform reconstruction	0.7646 ±0.0063	0.7173 ±0.0026	0.695 ±0.013	0.622 ±0.006	-3.2	-2.0

test subjects). Downstream recordings may overlap TUEG; TUAB and TUEV come from the same corpus family, and we did not exclude their patients from pretraining. The efficiency comparisons in steps and model size are controlled, but those with external models, which differ in data, duration and architecture, are not, and pretraining uses a single 2,388-hour budget.

Theory. Sec. 4 assumes stationarity, Gaussianity and a state that acts only through the spectrum. Real EEG contains transients (K-complexes, spikes, blinks), heavy tails and cross-channel phase relations. Contextual dependence, posterior state uncertainty and the positive log floor limit the idealized noise calculations. The analysis motivates the target, but does not identify the mechanism of the observed efficiency or prove optimality.

8. Conclusion

SpecBraM shows that what an EEG foundation model predicts matters as much as how large it is. Predicting the log power of masked patches in the classical frequency bands, a physical quantity that summarizes spectral levels without requiring local phase prediction, makes pretraining efficient: with backbone, data and steps fixed it reaches the quality of waveform reconstruction with far fewer steps and a much smaller model, and with 2,388 hours of data it has higher point estimates than CBraMod, LaBraM and CSBrain on nearly all tasks and REVE on most tasks under an identical linear probe, with strong low-label results on sleep staging. We hope this encourages choosing the pretraining targets of physiological signals by what the signal means rather than by what other modalities predict.

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SpecBraM: Efficient EEG Foundation Model Pretraining by Predicting Masked Band Power

Supplementary Material

A. Idealized Analysis: Statements and Proofs

Let $\boldsymbol{x} = (x_0, \dots, x_{N-1})$ be one channel of a patch, $f_k = kf_s/N$, and consider $1 \le k \le K = \lfloor (N-1)/2 \rfloor$. Define $X_k = \sum_t x_t e^{-i2\pi kt/N}$, $I_k = |X_k|^2/N$, $\varphi_k = \arg X_k$.

Assumption 1 (stationary Gaussian; state acts through the spectrum)**.** *Given the state θ, $\boldsymbol{x}$ is a segment of a zero-mean weakly stationary Gaussian process with positive spectral density S_θ, continuous except at finitely many points and continuous at the f_k used.*

Lemma 1 (Brillinger 1981, Thm. 4.4.1)**.** *Under Assumption 1, as $N \to \infty$, for a fixed set of frequencies, $X_k/\sqrt{NS_\theta(f_k)}$ converges in distribution to independent standard circular complex Gaussians. Hence approximately $I_k \approx S_\theta(f_k)E_k$ with E_k i.i.d. $\mathrm{Exp}(1)$, and φ_k i.i.d. uniform on $[0, 2\pi)$, independent of $\{I_k\}$, with a distribution that does not depend on θ.*

The Whittle approximation [35] follows:

$$-\log p(\boldsymbol{x} \mid \theta) \approx \sum_{k=1}^{K} \left[\log S_\theta(f_k) + \frac{I_k}{S_\theta(f_k)} \right] + \text{const.} \quad (3)$$

Proposition 1 (phase is ancillary)**.** *In the Whittle model of (3), $(I_1, \dots, I_K)$ is sufficient for θ, and the distribution of $(\varphi_1, \dots, \varphi_K)$ does not depend on θ.*

Proof. The data enter (3) only through $\{I_k\}$ (Fisher–Neyman factorization); the phase distribution is given by Lemma 1. □

This is a statement about a patch's marginal spectral likelihood. It does not imply that its phase is independent of visible context. Both the magnitude and phase of a Fourier coefficient fluctuate conditional on the state; correlations with neighbouring samples or other channels can make part of that realization predictable.

Assumption 2 (approximately flat bands; in-band observation)**.** *Let $\mathcal{F} = [0.5, 45]$ Hz contain disjoint bands $B_1, \dots, B_5$, possibly separated by transition regions. (i) For $f_k \in B_b$, $S_\theta(f_k) \approx s_b(\theta)$. (ii) The observation consists only of the in-band coefficients $\{X_k : f_k \in \bigcup_b B_b\}$ (equivalently, the signal is first band-passed to $\bigcup_b B_b$). Let $m_b = \#\{k : f_k \in B_b\}$.*

Part (ii) avoids any conflict with the continuity in Assumption 1: the spectrum may vary continuously everywhere; (i) only requires it to be nearly constant within each band.

Proposition 2 (band power in the restricted Whittle model)**.** *Under Assumption 2 with equality in (i), in the Whittle model of (3), $P_b = m_b^{-1} \sum_{f_k \in B_b} I_k$ is sufficient for $\boldsymbol{s} = (s_b)_b$, $\hat{s}_b = P_b$ is the maximum-likelihood estimate, and $\log P_b$ is the MLE of $\log s_b$.*

Proof. By (ii) the likelihood involves only in-band bins; by (i), $-\log p(\boldsymbol{x} \mid \boldsymbol{s}) \approx \sum_b [m_b \log s_b + m_b P_b / s_b] + \text{const}$. The data enter only through P_b; setting the derivative in s_b to zero gives $\hat{s}_b = P_b$. □

Remark. Prop. 2 is exact within this restricted likelihood; it is an idealization for non-flat spectra and finite EEG patches. It does not establish sufficiency for spectral information outside the observed bands or for arbitrary downstream labels. For an ideal circular band-pass $\tilde{h}_b$, Parseval gives $\frac{1}{N}\sum_t ((\tilde{h}_b * \boldsymbol{x})_t)^2 = \frac{2m_b}{N} P_b$, a per-band constant factor. The main target (Eq. (1), with $N{=}P$) uses $\log(a_b P_b + \varepsilon)$, where $a_b = 2m_b/N$. At zero floor this is $\log P_b + \log a_b$. At positive floor the transformation remains invertible, but its noise distribution is no longer a constant shift of the zero-floor log distribution.

Proposition 3 (conditional noise of zero-floor log power)**.** *In the independent-bin Whittle model with exactly flat bands, $P_b = s_b(\theta) G_b$ conditional on θ, where $G_b \sim \mathrm{Gamma}(m_b, 1/m_b)$ (shape, scale). Thus*

$$\begin{aligned} \mathrm{Var}[P_b \mid \theta] &= s_b(\theta)^2 / m_b, \\ \mathbb{E}[\log P_b \mid \theta] &= \log s_b(\theta) + \psi(m_b) - \log m_b, \\ \mathrm{Var}[\log P_b \mid \theta] &= \psi'(m_b) \approx 1/m_b. \end{aligned}$$

Proof. A sum of m i.i.d. $\mathrm{Exp}(1)$ is $\mathrm{Gamma}(m, 1)$, whose logarithm has mean $\psi(m)$ and variance $\psi'(m)$. □

For Eq. (1), the additional term $\log(G_b + \varepsilon/[a_b s_b(\theta)])$ generally has a state-dependent mean and variance. The zero-floor identities are therefore reference calculations, not exact noise identities for the implemented target; no negligible-floor approximation is assumed for all observations.

Assumption 3 (shared state and independent-patch idealization)**.** *The context $\boldsymbol{c}$ and the masked patch share the state θ, and given θ the target of the masked patch is independent of $\boldsymbol{c}$.*

This is an additional idealization used below for band power, not a consequence of phase ancillarity or patch-local target computation. Adjacent patches and other channels

may remain correlated conditional on the state. The correlated waveform model below does not impose this independence assumption.

Proposition 4 (risk decomposition for any target)**.** *For a square-integrable target Y, let $g(\boldsymbol{c}) = \mathbb{E}[Y \mid \boldsymbol{c}]$ and $V(\boldsymbol{c}) = \mathrm{Cov}(Y \mid \boldsymbol{c})$. Under squared Euclidean loss,*

$$R(f) = \mathbb{E}\|f(\boldsymbol{c}) - Y\|^2,$$
$$R^* = \inf_f R(f) = \mathbb{E}\,\mathrm{tr}\,V(\boldsymbol{c}),$$
$$R(f) - R^* = \mathbb{E}\|f(\boldsymbol{c}) - g(\boldsymbol{c})\|^2.$$

For scalar Y under Assumption 3, writing $\mu(\theta) = \mathbb{E}[Y \mid \theta]$ and $\sigma^2(\theta) = \mathrm{Var}(Y \mid \theta)$ gives $R^* = \mathbb{E}[\mathrm{Var}(\mu(\theta) \mid \boldsymbol{c})] + \mathbb{E}[\sigma^2(\theta)]$.

Proof. Expanding $f - Y = (f - g) + (g - Y)$, the cross term vanishes after conditioning on $\boldsymbol{c}$. The last identity follows from conditional independence: $\mathbb{E}[Y \mid \boldsymbol{c}] = \mathbb{E}[\mu(\theta) \mid \boldsymbol{c}]$ and $\mathrm{Var}(Y \mid \boldsymbol{c}) = \mathrm{Var}(\mu(\theta) \mid \boldsymbol{c}) + \mathbb{E}[\sigma^2(\theta) \mid \boldsymbol{c}]$; take expectations over $\boldsymbol{c}$. □

For zero-floor $Y_b = \log P_b$ under Assumption 3 and the exact flat-band Whittle model, let $k_b = \psi(m_b) - \log m_b$. Then

$$g_b(\boldsymbol{c}) = \mathbb{E}[\log s_b(\theta) \mid \boldsymbol{c}] + k_b,$$
$$V_b(\boldsymbol{c}) = \psi'(m_b) + \mathrm{Var}(\log s_b(\theta) \mid \boldsymbol{c}).$$

Both terms in the Bayes risk are irreducible for a fixed observed context. Only excess prediction error above this risk is reducible by improving f. Predicting a posterior average of spectral levels need not identify θ or a downstream label.

A separate correlated waveform model. Applying the independence assumption to a zero-mean waveform would give $g(\boldsymbol{c}) = \mathbf{0}$. Instead, for a fixed mask suppose $(\boldsymbol{x}_M, \boldsymbol{c})$ is jointly zero-mean Gaussian conditional on θ, allowing nonzero cross-covariance. Its conditional mean and innovation covariance are

$$\boldsymbol{m}_\theta(\boldsymbol{c}) = \Sigma_{Mc,\theta}\Sigma_{cc,\theta}^{-1}\boldsymbol{c},$$
$$Q_\theta = \Sigma_{MM,\theta} - \Sigma_{Mc,\theta}\Sigma_{cc,\theta}^{-1}\Sigma_{cM,\theta},$$

assuming $\Sigma_{cc,\theta}$ is invertible. With the state unknown, total expectation and covariance give

$$g_{\mathrm{raw}}(\boldsymbol{c}) = \mathbb{E}[\boldsymbol{m}_\theta(\boldsymbol{c}) \mid \boldsymbol{c}],$$
$$V_{\mathrm{raw}}(\boldsymbol{c}) = \mathbb{E}[Q_\theta \mid \boldsymbol{c}] + \mathrm{Cov}(\boldsymbol{m}_\theta(\boldsymbol{c}) \mid \boldsymbol{c}).$$

Thus the known-state innovation alone omits posterior state uncertainty. For per-coordinate waveform MSE, the Bayes risk is $\mathbb{E}\,\mathrm{tr}\,V_{\mathrm{raw}}(\boldsymbol{c})/|M|$.

Proposition 5 (the irreducible part adds gradient noise, not signal)**.** *Let $Y = g(\boldsymbol{c}) + \xi$ with $g(\boldsymbol{c}) = \mathbb{E}[Y \mid \boldsymbol{c}]$, $\mathbb{E}[\xi \mid \boldsymbol{c}] = 0$ and $\mathrm{Cov}(\xi \mid \boldsymbol{c}) = \Sigma(\boldsymbol{c})$. For a model f_w with Jacobian $J = \partial f_w(\boldsymbol{c})/\partial w$ and loss $\ell = \frac{1}{2}\|f_w(\boldsymbol{c}) - Y\|^2$, the stochastic gradient $G = J^\top(f_w(\boldsymbol{c}) - Y)$ satisfies $\mathbb{E}[G] = \nabla_w \frac{1}{2}\mathbb{E}\|f_w(\boldsymbol{c}) - g(\boldsymbol{c})\|^2$ and $\mathrm{Cov}(G) = \mathrm{Cov}\big(J^\top(f_w - g)\big) + \mathbb{E}\big[J^\top\Sigma(\boldsymbol{c})J\big]$.*

Proof. $G = J^\top(f_w - g) - J^\top\xi$. Conditioning on $\boldsymbol{c}$, $\mathbb{E}[J^\top\xi \mid \boldsymbol{c}] = 0$, so the second term has mean zero and is uncorrelated with the first; its covariance is $\mathbb{E}[J^\top\mathbb{E}[\xi\xi^\top \mid \boldsymbol{c}]J]$. □

The expected gradient depends only on the excess prediction error $f_w - g$; the conditional residual contributes $\mathbb{E}[J^\top\Sigma J]$. Here $\Sigma(\boldsymbol{c}) = V(\boldsymbol{c})$ is the *full* conditional covariance. For the vector of zero-floor log band powers under the independent-patch, flat-band Whittle model, let $\boldsymbol{\lambda}(\theta) = (\log s_b(\theta))_{b=1}^5$. Then

$$\Sigma_{\mathrm{BP}}(\boldsymbol{c}) = \mathrm{diag}_b\{\psi'(m_b)\} + \mathrm{Cov}(\boldsymbol{\lambda}(\theta) \mid \boldsymbol{c}).$$

For the correlated waveform model it is $V_{\mathrm{raw}}(\boldsymbol{c})$ above. The positive floor changes the band-power noise identities, and neither target's gradient covariance is determined by its marginal target variance alone.

For an L-smooth, lower-bounded population risk $\mathcal{R}(w)$, unbiased stochastic gradients with variance bounded by σ_G^2, independent mini-batch samples of size B, and suitable step sizes, a standard upper bound for returning an iterate with $\mathbb{E}\|\nabla_w\mathcal{R}\|^2 \le \epsilon$ has order $L\Delta/\epsilon + L\Delta\sigma_G^2/(B\epsilon^2)$, where Δ is the initial risk gap [13]. This upper bound can worsen with gradient variance; it does not establish that one target necessarily needs proportionally more steps, or bound steps to a downstream score. Comparing the targets also changes g, J, loss scale and smoothness. The gradient identity therefore motivates a possible mechanism rather than proving the observed speedup.

Empirical comparison. On TUAB evaluation windows (Table 3) the band-power model's error at epoch 16 is $0.16/0.12/0.10/0.06/0.08$ for $\delta/\theta/\alpha/\beta/\gamma$, against an idealized noise $\psi'(m_b)$ of $0.40/0.40/0.28/0.08/0.09$ (Table 7). Errors below these reference values show that the idealized calculation is not a quantitative noise floor for this setting. Contextual dependence, non-flat spectra, finite patches and the positive floor can all affect the comparison; these measurements do not isolate a cause or establish a general bound on Bayes risk. The waveform model explains 3% of its target's variance. Achieved R^2 measures the trained predictor, not the maximum predictability of either target.

B. Downstream Data Details

Beyond-sleep tasks. For MA, Mumtaz and PhysioNet-MI we reproduce CBraMod's `preprocessing_*.py`

Table 7. Ideal rectangular bands at our patch length (N=200, f_s=256 Hz, Δf=1.28 Hz): number of Fourier bins and idealized conditional sampling noise of zero-floor $\log P_b$ (Prop. 3). These are the bands and bin counts of the main target; the noise values are not measured conditional variances for Eq. (1).

Band	Range (Hz)	m_b	$\mathrm{sd}[\log P_b \mid \theta] = \sqrt{\psi'(m_b)}$
δ	0.5–4	3	0.63
θ	4–8	3	0.63
α	8–13	4	0.53
β	13–30	13	0.28
γ	30–45	12	0.29

scripts line by line (file ordering, split slices, channel selection, filtering, resampling to 200 Hz and window length) and store the samples once; every model is probed on these samples. FACED's official processed release was not reachable from our region; we rebuilt it from the public raw BIDS recordings following the FACED paper (last 30 s of each clip, 0.05–47 Hz, common average reference, 250 Hz, bad-channel interpolation; no ICA) and then applied CBraMod's FACED preprocessing. Our input convention adds, for our model only, 0.5–45 Hz band-pass, resampling to 256 Hz and per-recording robust scaling of the same samples, restricted to the 19 standard leads, which all four tasks contain.

C. Detailed Results per Dataset

Table 1 reports balanced accuracy only. Here every dataset is described and given its own table with the metrics that prior EEG foundation models report: balanced accuracy, Cohen's κ and weighted F1 for every task, and additionally AUROC and AUC-PR for the binary tasks. All numbers come from the strict linear probe of Sec. 5, measured by us with the released weights on identical samples and splits; balanced accuracy on the validation split is the only selection criterion (the regularization C and, for REVE, the readout).

Uncertainty. The external models are released as single checkpoints, so we evaluate one checkpoint of SpecBraM as well, our main released model, and quantify the uncertainty of every model in the same way. For mixed-label or multiclass tasks we use ordinary cluster resampling of test subjects with replacement. For binary tasks whose subjects each have one true label, subjects are resampled within true-label strata. For TUEV the cluster unit is the recording, since its evaluation files are anonymized. Candidate resamples missing any test-set class are discarded until 1,000 valid replicates are obtained. Every metric is recomputed on the pooled samples of the drawn clusters; the 2.5th and 97.5th percentiles form the nominal 95% confidence interval. These intervals describe test-cohort sampling variation conditional on the fitted checkpoint and probe, not pretraining-seed variability. To show the variation of our pretraining, each table also lists SpecBraM's mean and standard deviation over its three pretraining seeds; this row is not used for the comparison.

Confusion matrices. For every dataset we also show the row-normalized confusion matrices of SpecBraM and of REVE, the strongest baseline, and their difference, to see which classes each model recognizes.

Paired differences. Overlapping intervals do not imply that two models are equivalent, so we also bootstrap the difference itself: on each of the 1,000 resamples the score of every external model is subtracted from SpecBraM's on the same drawn clusters, preserving the pairing (Table 8). The generated table gives the nominal, unadjusted 95% intervals for each difference; no correction for multiple comparisons is applied. An interval excluding zero is evidence of a difference under this procedure. An interval containing zero indicates no detected difference, not equivalence or noninferiority. The intervals condition on the evaluated checkpoints and do not include training variability.

Per-subject comparison. A higher pooled score could come from a few subjects. Fig. 4 compares SpecBraM and REVE subject by subject, and Table 9 counts the subjects on which SpecBraM is better, equal or worse than each external model. SpecBraM is better than CBraMod and CSBrain on every sleep subject and than LaBraM and BIOT on all but one subject per sleep dataset; on every other dataset it is better than CBraMod, LaBraM and CSBrain on more subjects than it is worse. Against REVE it is better on 81% of the HMC subjects and on three of the four MA subjects, and about even on ISRUC, Mumtaz, FACED and PhysioNet-MI. On the clinical corpora the comparison is close in both directions: REVE is better on slightly more TUAB patients (41% vs. 36%, 23% ties) and SpecBraM on slightly more TUEV recordings (45% vs. 37%).

C.1. Sleep Staging: ISRUC

ISRUC subgroup I [19] contains one overnight polysomnography of each of 100 adults with sleep complaints, scored by experts in 30-s epochs into five AASM stages (W, N1, N2, N3, REM). We use the six mastoid-referenced EEG leads (F3-A2, C3-A2, O1-A2, F4-A1, C4-A1, O2-A1; 200 Hz), of which F3/F4 are missing in some recordings, and the subject split of CBraMod and CSBrain (subjects 1–80/81–90/91–100; 71,936/9,438/8,813 epochs). A 30-s epoch is represented by the mean of the features of its three 10-s windows. Fig. 5: The two models make the same kinds of errors: N1 is confused with N2 and REM, and N3 with N2. Per-class recall differs by at most 2.4 points, consistent with the small difference in balanced-accuracy point estimates.

Table 8. **Paired bootstrap of the difference** in balanced accuracy (points) between SpecBraM's main checkpoint and each external model, computed on the same 1,000 resamples of the test subjects (TUEV: recordings), with its nominal, unadjusted 95% confidence interval. Positive: SpecBraM better. Bold: the interval excludes zero; this is not an equivalence test.

Dataset	CBraMod	LaBraM-Base	BIOT	CSBrain	REVE-Base
ISRUC	**+6.2** [+4.28, +8.01]	**+3.8** [+1.93, +5.41]	**+5.4** [+2.81, +8.11]	**+6.0** [+4.85, +7.11]	-0.3 [-1.89, +1.36]
HMC	**+12.3** [+9.53, +15.51]	**+11.0** [+8.45, +13.63]	**+8.4** [+6.06, +10.89]	**+13.1** [+9.82, +16.35]	**+3.8** [+1.83, +5.90]
TUAB	**+3.5** [+1.09, +6.12]	**+3.7** [+1.52, +6.06]	–	**+4.1** [+1.63, +6.44]	-0.8 [-2.71, +1.21]
TUEV	+3.0 [-3.06, +9.68]	**+8.8** [+1.87, +15.76]	–	-0.9 [-7.48, +10.32]	+0.9 [-4.63, +10.22]
MA	+12.8 [-3.12, +25.00]	**+10.1** [+5.21, +13.19]	–	**+22.6** [+12.50, +32.29]	+9.7 [+0.00, +17.36]
Mumtaz	**+4.7** [+0.08, +11.37]	**+6.3** [+0.90, +12.38]	–	**+7.6** [+3.30, +12.04]	+3.7 [-6.65, +17.43]
FACED	**+5.0** [+2.48, +7.52]	**+5.3** [+3.18, +7.58]	–	**+4.1** [+0.04, +8.01]	+2.1 [-0.70, +4.58]
PhysioNet-MI	**+10.5** [+6.49, +14.78]	**+11.8** [+8.70, +14.77]	–	**+11.9** [+7.68, +16.51]	+1.4 [-2.03, +5.15]

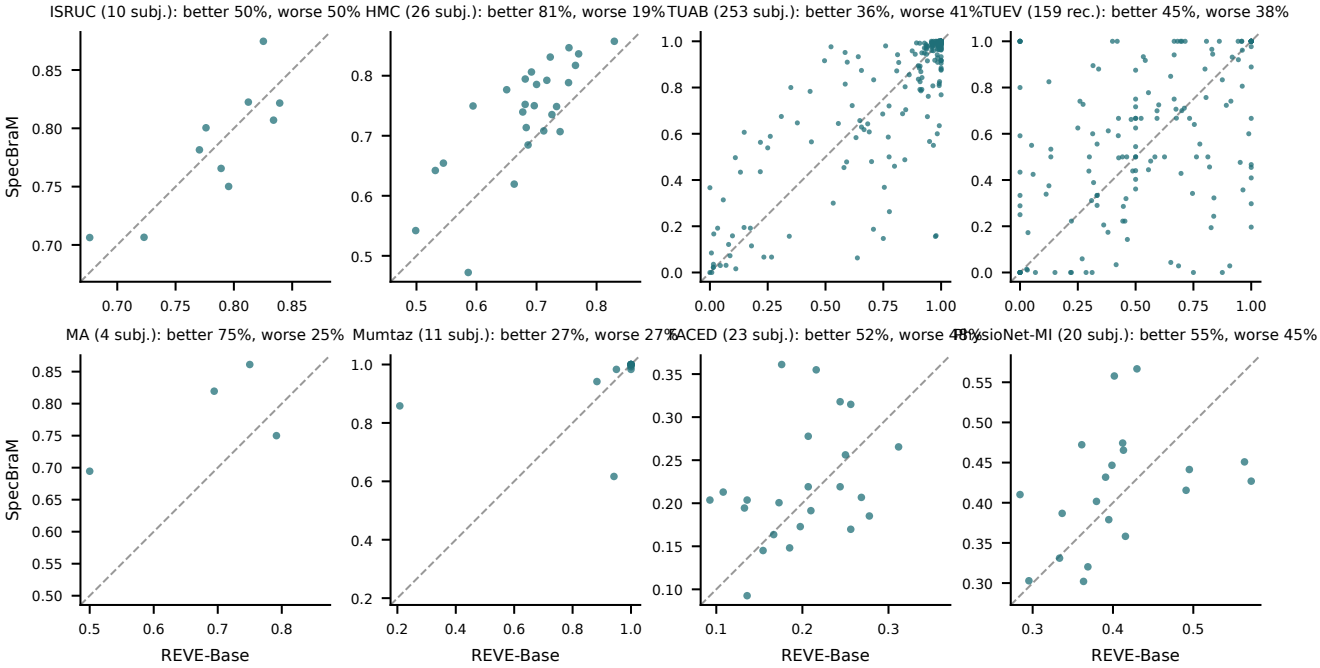


Figure 4. **Per-subject balanced accuracy**, SpecBraM's main checkpoint vs. REVE-Base (strict linear probe; one point per test subject, TUEV: per recording; dashed: equality). Titles: share of subjects on which SpecBraM is better / worse.

Table 9. **Per-subject comparison**: percentage of test subjects (TUEV: recordings) on which SpecBraM's balanced accuracy is higher / equal / lower than that of each external model (strict linear probe, one checkpoint per model).

Dataset	CBraMod	LaBraM-Base	BIOT	CSBrain	REVE-Base
ISRUC (10 subj.)	100/0/0	90/0/10	90/0/10	100/0/0	50/0/50
HMC (26 subj.)	100/0/0	96/0/4	96/0/4	100/0/0	81/0/19
TUAB (253 subj.)	60/13/27	65/10/25	–	62/11/27	36/23/41
TUEV (159 rec.)	50/17/33	50/19/31	–	45/21/34	45/18/37
MA (4 subj.)	75/0/25	100/0/0	–	100/0/0	75/0/25
Mumtaz (11 subj.)	45/36/19	64/27/9	–	82/18/0	27/45/28
FACED (23 subj.)	78/0/22	83/0/17	–	65/0/35	52/0/48
PhysioNet-MI (20 subj.)	85/0/15	95/0/5	–	90/0/10	55/0/45

Table 10. **Detailed results on ISRUC** (strict linear probe). One checkpoint per model, with the 95% confidence interval from 1,000 bootstrap resamples of the 10 test subjects. Below SpecBraM's main checkpoint: mean±std over its three pretraining seeds. Bold: best per column.

Model	Balanced Acc.	Cohen's κ	Weighted F1
SpecBraM	0.790 [0.764, 0.816]	0.733 [0.692, 0.773]	**0.795** [0.757, 0.830]
three seeds, mean±std	0.797 ±0.006	0.744 ±0.010	0.803 ±0.008
CBraMod	0.728 [0.702, 0.752]	0.658 [0.608, 0.705]	0.734 [0.698, 0.770]
LaBraM-Base	0.752 [0.720, 0.777]	0.686 [0.642, 0.724]	0.754 [0.716, 0.787]
BIOT	0.736 [0.691, 0.771]	0.666 [0.601, 0.715]	0.740 [0.686, 0.785]
CSBrain	0.730 [0.706, 0.752]	0.674 [0.638, 0.714]	0.749 [0.719, 0.782]
REVE-Base	**0.793** [0.765, 0.816]	**0.736** [0.703, 0.769]	**0.795** [0.765, 0.824]

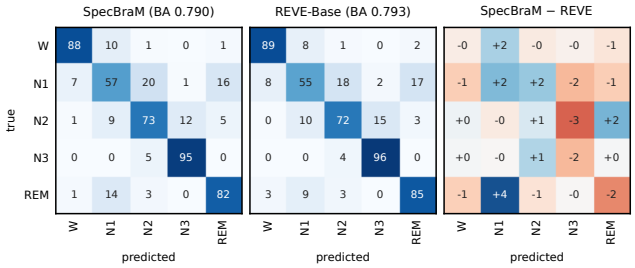


Figure 5. **Confusion matrices on ISRUC** (strict linear probe, row-normalized recall in %): SpecBraM's main checkpoint, REVE-Base and their difference (blue: SpecBraM higher).

C.2. Sleep Staging: HMC

HMC [1] contains 151 clinical polysomnographies from the Haaglanden Medisch Centrum, scored in 30-s epochs into the same five stages. We use the four EEG leads (F4-M1, C4-M1, O2-M1, C3-M2; 256 Hz) and the split of CBraMod and CSBrain by sorted subject ID (100/25/26 recordings; 91,248/22,124/23,871 epochs). Epochs are represented as for ISRUC. Fig. 6: SpecBraM's lead over REVE comes from the transitional and REM stages: its recall is higher by 5.0 points for N1, 6.0 for N2 and 7.7 for REM, while W and N3, the stages with the most distinct spectra, are recognized equally well by both.

Table 11. **Detailed results on HMC** (strict linear probe). One checkpoint per model, with the 95% confidence interval from 1,000 bootstrap resamples of the 26 test subjects. Below SpecBraM's main checkpoint: mean±std over its three pretraining seeds. Bold: best per column.

Model	Balanced Acc.	Cohen's κ	Weighted F1
SpecBraM	**0.742** [0.699, 0.776]	**0.675** [0.620, 0.724]	**0.751** [0.707, 0.791]
three seeds, mean±std	0.743 ±0.002	0.675 ±0.002	0.749 ±0.003
CBraMod	0.619 [0.576, 0.654]	0.513 [0.459, 0.567]	0.619 [0.572, 0.665]
LaBraM-Base	0.632 [0.583, 0.673]	0.530 [0.473, 0.585]	0.636 [0.586, 0.680]
BIOT	0.658 [0.609, 0.697]	0.562 [0.507, 0.615]	0.661 [0.616, 0.704]
CSBrain	0.611 [0.580, 0.641]	0.515 [0.472, 0.554]	0.625 [0.590, 0.660]
REVE-Base	0.704 [0.664, 0.737]	0.623 [0.577, 0.670]	0.707 [0.671, 0.745]

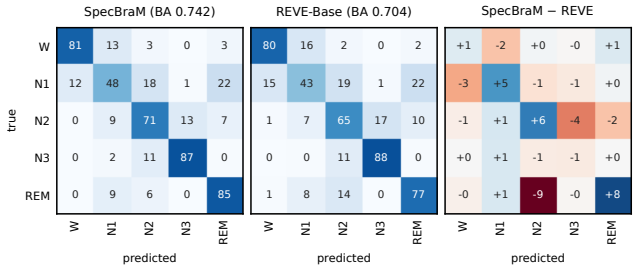


Figure 6. **Confusion matrices on HMC** (strict linear probe, row-normalized recall in %): SpecBraM's main checkpoint, REVE-Base and their difference (blue: SpecBraM higher).

Case study: one night per dataset. Fig. 7 shows the hypnograms of one test subject per sleep dataset, chosen by rule as the subject whose SpecBraM balanced accuracy is closest to the median of its dataset, with the stages predicted by SpecBraM, REVE and CBraMod from the frozen features of each 30-s epoch alone (no sequence model). All models follow the sleep cycles; their errors concentrate at stage transitions and in N1. On the ISRUC night the three models are close (balanced accuracy 0.800, 0.776 and 0.728). On the HMC night SpecBraM reaches 0.750 against 0.594 for REVE, which scores almost a third of the N2 epochs as N3 (99 of 319 vs. 27 for SpecBraM) and half of the REM epochs as N2 (69 of 132 vs. 12), the same er-

rors that dominate REVE's confusion matrix on the whole dataset (Fig. 6).

C.3. Abnormal EEG Detection: TUAB

TUAB [25] contains clinical EEG recordings of the Temple University Hospital labeled normal or abnormal by neurologists. Following CBraMod, recordings are resampled to 200 Hz, filtered (0.3–75 Hz, 60 Hz notch), converted to 16 bipolar channels of the TCP montage and cut into non-overlapping 10-s windows that inherit the recording label. The official evaluation folder is the test set; the training folder is split 80/20 by sorted patient ID within each class (297,103/75,407/36,945 windows). The prepared metadata show three patients shared between training and validation under this original per-class split. Test patients are disjoint from both training and validation. All models use these same samples and splits; the overlap limits validation independence. This downstream split issue is separate from possible TUEG pretraining overlap. Fig. 8: Both models recognize normal and abnormal recordings almost equally well (recall within one point of each other); REVE's slightly higher balanced accuracy is spread over both classes.

C.4. Event Classification: TUEV

TUEV [25] annotates clinical EEG with six event classes: spike and sharp wave (SPSW), generalized and lateralized periodic epileptiform discharges (GPED, PLED), eye movement (EYEM), artifact (ARTF) and background (BCKG). Following CBraMod, each annotated 1-s event is represented by a 5-s window centred on it, with the same filtering and 16-channel bipolar montage as TUAB. The official evaluation folder is the test set and the training folder is split 80/20 by sorted patient ID (68,445/15,487/29,421 windows), with no patient overlap among the three downstream splits. Background makes up 65% of the windows; Cohen's κ and weighted F1 are dominated by it, whereas balanced accuracy weights the rare epileptiform classes equally. This explains why REVE has the highest κ and weighted F1 here although its balanced accuracy is below SpecBraM's and CSBrain's. Fig. 9: The two models differ in kind. SpecBraM detects spike and sharp waves (SPSW recall 25% vs. 0%; REVE never predicts this class) and eye movements (73% vs. 48%), events with a distinct spectral signature. REVE is better on the periodic discharges (GPED 90% vs. 61%, PLED 47% vs. 32%), which involve waveform shape and repetition over seconds. This pattern is consistent with P4, but the models also differ in architecture and pretraining data, so it does not isolate the effect of phase invariance.

Examples. Fig. 10 shows four test windows on which the two models disagree, chosen by rule: for each class, the window that the correct model classifies with the highest confidence. SpecBraM recognizes a spike and sharp wave that REVE labels as a lateralized periodic discharge, and an eye-movement window that REVE labels as background. REVE recognizes a generalized and a lateralized periodic discharge that SpecBraM labels as background. In the GPED window the discharges recur about every two seconds across all channels. Their waveform and repetition extend beyond the band power of a single 0.78-s patch. The patch-local target does not directly supervise this longer temporal pattern (P4), although the encoder can still represent temporal and waveform information.

C.5. Mental Workload: MA

The mental arithmetic dataset [40] records 36 subjects at rest and during serial subtraction with 20 leads (19 standard 10–20 leads and A2-A1; 500 Hz). Following CBraMod, recordings are resampled to 200 Hz and cut into 5-s windows labeled rest or task, and the sorted subjects are split 28/4/4 (1,343/172/192 windows). With four test subjects, MA has by far the widest confidence intervals of all tasks. Fig. 11: SpecBraM is better on both conditions (rest +9.0, task +10.4 points recall), so its lead is not a shift of the decision threshold.

C.6. Depression: Mumtaz

Mumtaz [24] contains resting-state EEG (eyes open and closed) of patients with major depressive disorder and healthy controls, recorded with 19 linked-ear-referenced leads at 256 Hz. Following CBraMod, recordings are resampled to 200 Hz, filtered (0.3–75 Hz, 50 Hz notch) and cut into 5-s windows that inherit the subject's diagnosis; recordings are split by subject within each group (4,891/1,041/1,211 windows). Fig. 12: SpecBraM's gain comes from fewer healthy subjects classified as depressed (recall of the healthy class +12.6 points), at a cost of 5.2 points on the depressed class.

C.7. Emotion Recognition: FACED

FACED [9] records 123 subjects watching 28 video clips that elicit nine emotion categories (anger, disgust, fear, sadness, neutral, amusement, inspiration, joy, tenderness), with 32 leads at 250 Hz. The last 30 s of each clip are resampled to 200 Hz and cut into three 10-s windows labeled with the clip's category; the sorted subjects are split 80/20/23 (6,720/1,680/1,932 windows; chance level 1/9). Fig. 13: Both models are far from perfect on nine emotions. SpecBraM's gain is concentrated on neutral (+11.2 points) and tenderness (+11.6), the two categories at the low-arousal end, while the other categories differ by a few points in either direction.

C.8. Motor Imagery: PhysioNet-MI

PhysioNet-MI [27] records 109 subjects with 64 leads at 160 Hz while they imagine opening and closing the left fist,

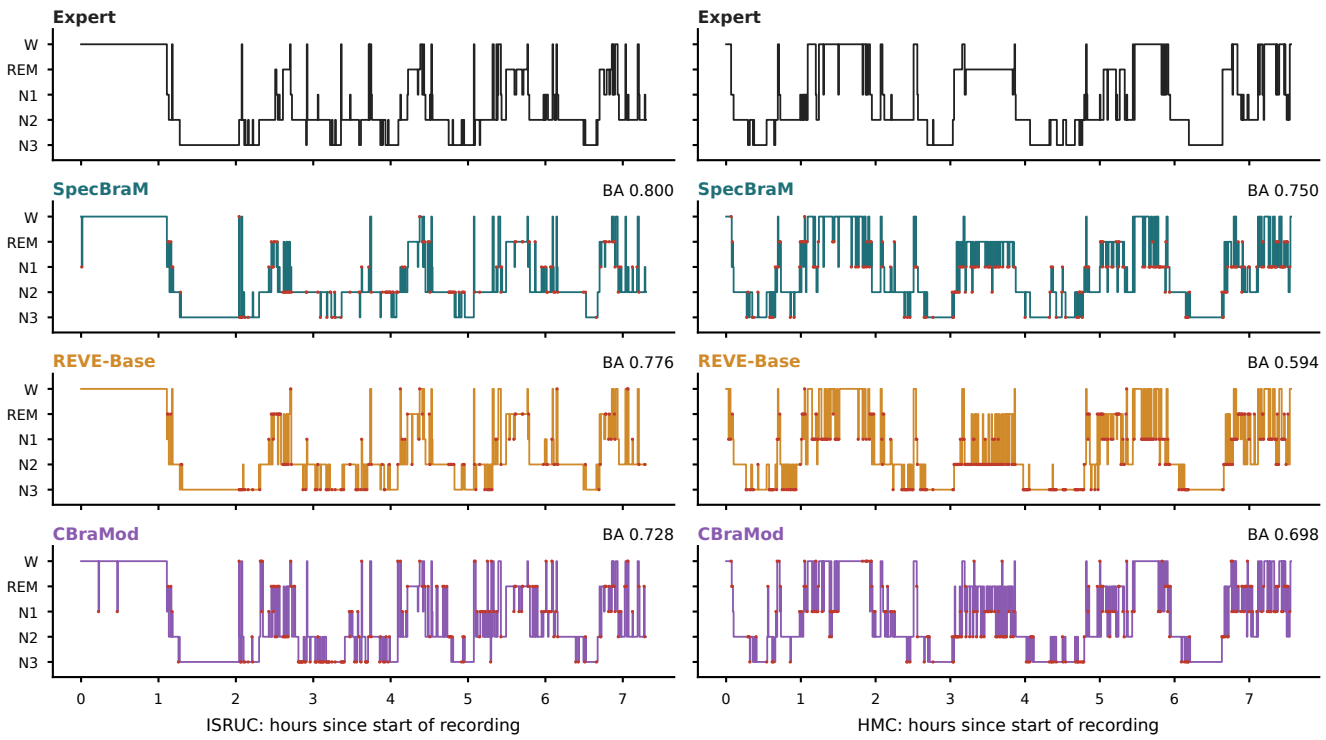


Figure 7. **Whole-night hypnograms** of one test subject per sleep dataset (left: ISRUC, right: HMC; the subject whose SpecBraM balanced accuracy is closest to the dataset median). Top: expert scoring; below: stages predicted by the strict linear probe on each model's frozen features, epoch by epoch. Red dots: errors.

Table 12. **Detailed results on TUAB** (strict linear probe). One checkpoint per model, with the 95% confidence interval from 1,000 bootstrap resamples of the 253 test subjects. Below SpecBraM's main checkpoint: mean±std over its three pretraining seeds. Bold: best per column.

Model	Balanced Acc.	Cohen's κ	Weighted F1	AUROC	AUC-PR
SpecBraM	0.813 [0.775, 0.850]	0.627 [0.554, 0.700]	0.815 [0.778, 0.851]	**0.891** [0.851, 0.924]	0.897 [0.860, 0.929]
three seeds, mean±std	0.805 ±0.007	0.611 ±0.014	0.807 ±0.007	0.880 ±0.011	0.890 ±0.008
CBraMod	0.778 [0.738, 0.814]	0.558 [0.480, 0.629]	0.780 [0.742, 0.816]	0.856 [0.810, 0.894]	0.857 [0.812, 0.893]
LaBraM-Base	0.776 [0.740, 0.812]	0.554 [0.484, 0.624]	0.779 [0.744, 0.814]	0.856 [0.817, 0.891]	0.851 [0.806, 0.889]
CSBrain	0.772 [0.736, 0.805]	0.546 [0.475, 0.611]	0.774 [0.740, 0.807]	0.845 [0.804, 0.882]	0.836 [0.786, 0.878]
REVE-Base	**0.821** [0.781, 0.861]	**0.644** [0.565, 0.722]	**0.823** [0.784, 0.861]	0.889 [0.848, 0.925]	**0.902** [0.864, 0.933]

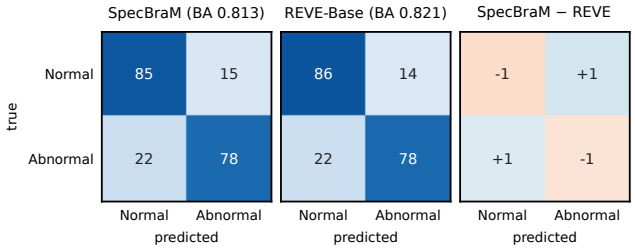


Figure 8. **Confusion matrices on TUAB** (strict linear probe, row-normalized recall in %): SpecBraM's main checkpoint, REVE-Base and their difference (blue: SpecBraM higher).

the right fist, both fists or both feet. Following CBraMod, the imagery runs are high-pass filtered at 0.3 Hz, notch filtered at 60 Hz and resampled to 200 Hz, and the 4-s window after each cue is labeled with the imagined movement; the sorted subjects are split 70/19/20 (6,300/1,734/1,803 windows; chance level $1/4$). Fig. 14: SpecBraM is better on bilateral imagery (both fists $+4.2$, both feet $+4.9$ points) and slightly worse on the left fist (-5.5); distinguishing the left and right fist requires lateralized spatial patterns that a mean-pooled readout dilutes.

Table 13. **Detailed results on TUEV** (strict linear probe). One checkpoint per model, with the 95% confidence interval from 1,000 bootstrap resamples of the 159 test recordings. Below SpecBraM's main checkpoint: mean±std over its three pretraining seeds. Bold: best per column.

Model	Balanced Acc.	Cohen's κ	Weighted F1
SpecBraM	0.555 [0.491, 0.648]	0.467 [0.369, 0.557]	0.722 [0.659, 0.777]
three seeds, mean±std	0.561 ±0.025	0.474 ±0.049	0.724 ±0.027
CBraMod	0.525 [0.471, 0.609]	0.399 [0.327, 0.480]	0.669 [0.598, 0.742]
LaBraM-Base	0.468 [0.416, 0.545]	0.327 [0.228, 0.423]	0.624 [0.518, 0.710]
CSBrain	**0.564** [0.505, 0.621]	0.421 [0.348, 0.489]	0.683 [0.619, 0.745]
REVE-Base	0.546 [0.500, 0.587]	**0.566** [0.439, 0.655]	**0.778** [0.727, 0.820]

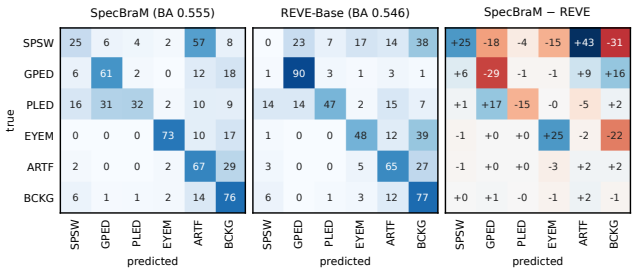


Figure 9. **Confusion matrices on TUEV** (strict linear probe, row-normalized recall in %): SpecBraM's main checkpoint, REVE-Base and their difference (blue: SpecBraM higher).

D. Band Power versus Waveform Reconstruction beyond Sleep

Table 18 extends the target study of Sec. 6.3 to the other tasks. Band power has higher mean scores on mental arithmetic (+6.2), depression screening (+1.8) and event classification (+2.8), and close mean scores on emotion recognition, motor imagery and abnormality detection; these comparisons do not establish equivalence.

E. Fine-Tuning

Sleep staging. Following CBraMod and CSBrain, 30-s epochs are grouped in sequences of 20; each epoch is encoded (mean of its three 10-s windows), and a shared one-layer transformer predicts one stage per epoch. Encoder and head are trained with AdamW (learning rate 10^{-4}, weight decay 0.05) for up to 50 epochs with early stopping on validation balanced accuracy (patience 10). SpecBraM reaches 0.8177±0.0032 balanced accuracy on ISRUC and 0.7696±0.0012 on HMC (Table 19), above the listed published point estimates (CSBrain 0.7925/0.7345, REVE 0.7819 on ISRUC), with the highest Cohen's κ on both datasets among the listed methods. The models differ in architecture, data and training recipe, and our standard deviations are across pretraining seeds, unlike the baselines' fine-tuning seeds. After fine-tuning, the gap to the matched raw-reconstruction model narrows to about 1.14/1.45 points (0.8062/0.7551). In these sleep experiments, the target differences are larger for frozen and low-label probes than after full fine-tuning.

Other tasks. Table 20 fine-tunes the encoder on the non-sleep tasks, for SpecBraM and the matched raw-reconstruction model (layer-wise learning-rate decay, early stopping on validation balanced accuracy). Band power has higher mean scores on mental arithmetic, motor imagery, abnormality detection and event classification, and the mean scores are close on depression screening and emotion recognition.

F. Variants of the Band-Power Target

Table 21 varies the spectral estimator while keeping the main model otherwise unchanged. With 0.78-s patches the δ and θ bands contain only three Fourier bins, motivating variance reduction (Table 7). A multitaper estimate (three DPSS tapers, $NW{=}2$) averages eigenspectra with ±2.6 Hz smoothing; the variance reduction of the integrated log-band target is not measured here. It gives nearby full-label mean scores and higher means with 1% of the training epochs. Both estimators support strong sleep representations, without identifying target-noise reduction as the cause.

G. Factors besides the Target

The comparison with external models (Sec. 6.1) also differs in model size, input convention and readout. Table 22 varies each in turn. *Readout*: averaging tokens within each channel and concatenating the channels, still followed by a logistic regression, raises CBraMod by 4.3/6.5 points and LaBraM by 1.2/6.3, while SpecBraM is unchanged on ISRUC and gains 0.7 on HMC; mean pooling therefore underreads these models, but SpecBraM still leads them by 2.5–6.6 points. REVE-Base reaches 0.7995 on ISRUC with this readout, close to SpecBraM's point estimate. *Input*: normalizing CBraMod's and LaBraM's input per recording, as for our model, helps CBraMod (+1.4/+5.4) and is mixed for LaBraM (−2.0/+2.0). *Scale*: at CBraMod's size (4.2M parameters) our band-power model loses only 0.8/1.5 points and still exceeds CBraMod by 6.0/10.9, while raw reconstruction loses 1.6/4.2 points. Model size thus explains little of the gap to CBraMod and LaBraM, and the target matters more at small scale.

H. Handcrafted Spectral Features

Since the target is a spectral feature, a natural question is what self-supervision adds over computing spectral features directly. Table 23 probes two handcrafted feature sets with the same protocol. The rich set is a strong reference: on sleep staging it exceeds CBraMod and LaBraM and the raw-reconstruction model on HMC, but SpecBraM exceeds it

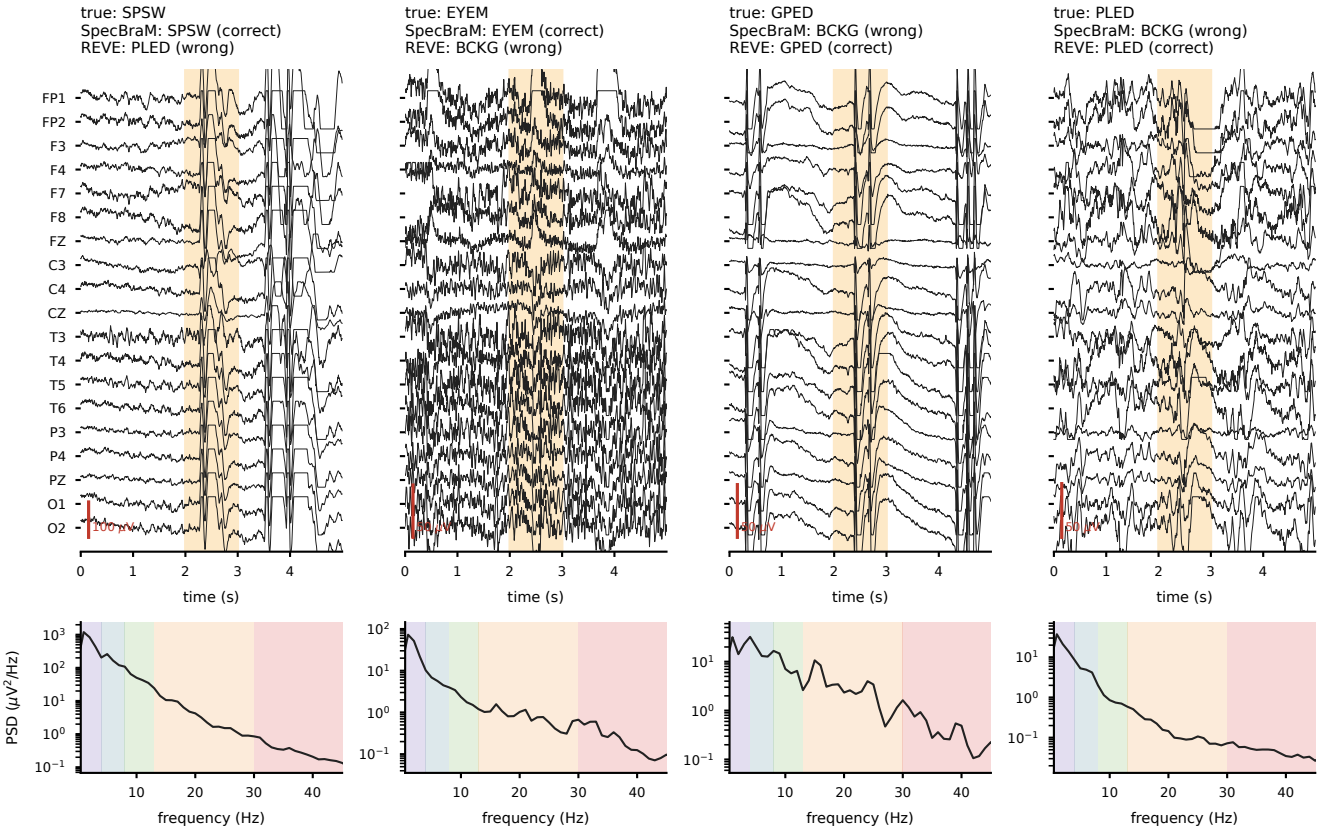


Figure 10. **TUEV examples on which SpecBraM and REVE disagree** (strict linear probe; for each class the window that the correct model classifies most confidently). Top: the 5-s window (19 monopolar leads, CBraMod's preprocessing; shaded: the annotated 1-s event; traces clipped to their row). Bottom: power spectral density averaged over channels, with the five bands of the target.

Table 14. **Detailed results on MA** (strict linear probe). One checkpoint per model, with the 95% confidence interval from 1,000 bootstrap resamples of the 4 test subjects. Below SpecBraM's main checkpoint: mean±std over its three pretraining seeds. Bold: best per column.

Model	Balanced Acc.	Cohen's κ	Weighted F1	AUROC	AUC-PR
SpecBraM	**0.781** [0.722, 0.840]	**0.583** [0.526, 0.655]	**0.846** [0.817, 0.876]	**0.912** [0.878, 0.981]	**0.783** [0.751, 0.942]
three seeds, mean±std	0.747 ±0.030	0.500 ±0.093	0.810 ±0.047	0.861 ±0.052	0.706 ±0.108
CBraMod	0.653 [0.507, 0.799]	0.256 [0.011, 0.510]	0.686 [0.525, 0.811]	0.708 [0.532, 0.884]	0.515 [0.305, 0.730]
LaBraM-Base	0.681 [0.597, 0.736]	0.342 [0.205, 0.452]	0.746 [0.700, 0.793]	0.734 [0.649, 0.834]	0.571 [0.432, 0.724]
CSBrain	0.556 [0.403, 0.653]	0.095 [-0.175, 0.275]	0.626 [0.543, 0.711]	0.586 [0.402, 0.747]	0.306 [0.218, 0.451]
REVE-Base	0.684 [0.562, 0.771]	0.361 [0.143, 0.551]	0.758 [0.679, 0.838]	0.757 [0.669, 0.924]	0.587 [0.485, 0.802]

by 3.3/2.1 points with all labels; with 1% of the labels the two are close (0.765/0.670 vs. 0.747/0.674). Beyond sleep, SpecBraM is clearly ahead on depression screening and the two clinical tasks and about level on emotion recognition and motor imagery. The handcrafted probes select the lower edge of the shared grid $C \in \{0.01, 0.1, 1\}$, so a wider grid might raise them slightly.

Table 15. **Detailed results on Mumtaz** (strict linear probe). One checkpoint per model, with the 95% confidence interval from 1,000 bootstrap resamples of the 11 test subjects. Below SpecBraM's main checkpoint: mean±std over its three pretraining seeds. Bold: best per column.

Model	Balanced Acc.	Cohen's κ	Weighted F1	AUROC	AUC-PR
SpecBraM	**0.939** [0.868, 0.995]	**0.879** [0.733, 0.991]	**0.940** [0.865, 0.995]	**0.991** [0.964, 1.000]	**0.991** [0.967, 1.000]
three seeds, mean±std	0.942 ±0.007	0.884 ±0.015	0.942 ±0.007	0.994 ±0.003	0.994 ±0.003
CBraMod	0.893 [0.760, 0.988]	0.787 [0.513, 0.978]	0.893 [0.744, 0.989]	0.973 [0.911, 1.000]	0.973 [0.908, 1.000]
LaBraM-Base	0.876 [0.744, 0.986]	0.754 [0.484, 0.974]	0.877 [0.735, 0.987]	0.963 [0.882, 0.999]	0.960 [0.874, 0.999]
CSBrain	0.864 [0.774, 0.943]	0.727 [0.549, 0.887]	0.864 [0.772, 0.944]	0.947 [0.878, 0.990]	0.942 [0.862, 0.987]
REVE-Base	0.902 [0.743, 0.994]	0.804 [0.481, 0.989]	0.901 [0.728, 0.995]	0.968 [0.879, 1.000]	0.963 [0.861, 1.000]

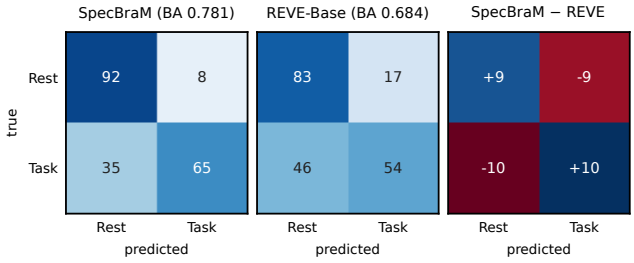


Figure 11. **Confusion matrices on MA** (strict linear probe, row-normalized recall in %): SpecBraM's main checkpoint, REVE-Base and their difference (blue: SpecBraM higher).

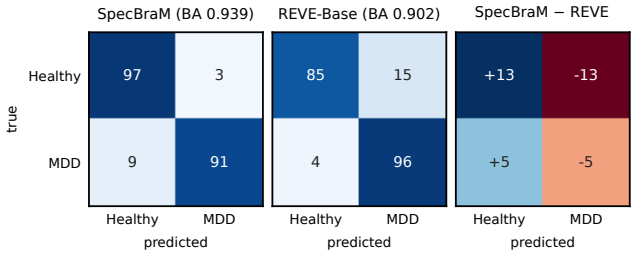


Figure 12. **Confusion matrices on Mumtaz** (strict linear probe, row-normalized recall in %): SpecBraM's main checkpoint, REVE-Base and their difference (blue: SpecBraM higher).

Table 16. **Detailed results on FACED** (strict linear probe). One checkpoint per model, with the 95% confidence interval from 1,000 bootstrap resamples of the 23 test subjects. Below SpecBraM's main checkpoint: mean±std over its three pretraining seeds. Bold: best per column.

Model	Balanced Acc.	Cohen's κ	Weighted F1
SpecBraM	**0.221** [0.195, 0.248]	**0.126** [0.096, 0.155]	**0.219** [0.193, 0.245]
three seeds, mean±std	0.216 ±0.013	0.119 ±0.015	0.214 ±0.010
CBraMod	0.171 [0.156, 0.188]	0.069 [0.052, 0.087]	0.167 [0.151, 0.182]
LaBraM-Base	0.167 [0.152, 0.181]	0.064 [0.047, 0.079]	0.163 [0.144, 0.177]
CSBrain	0.179 [0.161, 0.199]	0.077 [0.057, 0.098]	0.171 [0.153, 0.189]
REVE-Base	0.200 [0.179, 0.225]	0.101 [0.076, 0.130]	0.200 [0.177, 0.224]

Table 17. **Detailed results on PhysioNet-MI** (strict linear probe). One checkpoint per model, with the 95% confidence interval from 1,000 bootstrap resamples of the 20 test subjects. Below SpecBraM's main checkpoint: mean±std over its three pretraining seeds. Bold: best per column.

Model	Balanced Acc.	Cohen's κ	Weighted F1
SpecBraM	**0.417** [0.384, 0.450]	**0.223** [0.179, 0.267]	**0.416** [0.383, 0.449]
three seeds, mean±std	0.397 ±0.018	0.196 ±0.024	0.396 ±0.018
CBraMod	0.312 [0.286, 0.340]	0.083 [0.048, 0.119]	0.310 [0.282, 0.337]
LaBraM-Base	0.299 [0.285, 0.314]	0.065 [0.047, 0.085]	0.299 [0.282, 0.312]
CSBrain	0.298 [0.273, 0.325]	0.064 [0.030, 0.100]	0.295 [0.270, 0.320]
REVE-Base	0.403 [0.371, 0.438]	0.204 [0.161, 0.250]	0.403 [0.371, 0.437]

Table 18. **Band power vs. raw-waveform reconstruction beyond sleep** (same architecture, data and steps; strict linear probe, balanced accuracy, mean±std over three pretraining seeds). Bold: best per column.

Target	TUAB	TUEV	MA	Mumtaz	FACED	PhysioNet-MI
Band power (ours)	**0.805** ±0.007	**0.561** ±0.025	**0.747** ±0.030	**0.942** ±0.007	0.216 ±0.013	**0.397** ±0.018
Raw-waveform reconstruction	0.804 ±0.002	0.533 ±0.027	0.685 ±0.051	0.924 ±0.013	**0.220** ±0.010	0.395 ±0.013

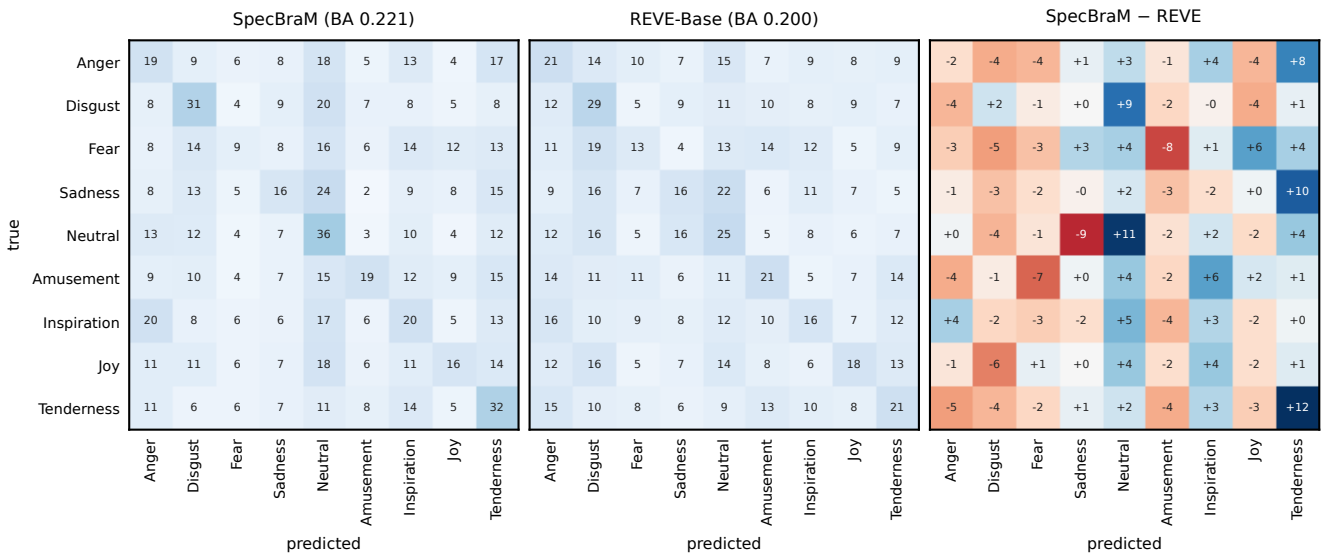


Figure 13. **Confusion matrices on FACED** (strict linear probe, row-normalized recall in %): SpecBraM's main checkpoint, REVE-Base and their difference (blue: SpecBraM higher).

Table 19. **Sleep staging, fine-tuned** (5-class, 30-s epochs in sequences of 20, one shared 1-layer sequence transformer; identical subject-disjoint splits as Zhou et al. [39]). Baselines as reported by Zhou et al. [39] (REVE: its paper, ISRUC only). Our rows: mean±std over three *pretraining* seeds (one fine-tuning run each), so the spread includes pretraining variance, unlike the baselines' fine-tuning-seed spread. Bold: best per column.

		ISRUC			HMC		
Method	Pretrain (h)	BA	κ	Weighted F1	BA	κ	Weighted F1
Task-specific models							
EEGNet [22]	–	0.7154 ±0.0121	0.7040 ±0.0173	0.7513 ±0.0124	0.6534 ±0.0122	0.5886 ±0.0201	0.6536 ±0.0168
EEGConformer [29]	–	0.7400 ±0.0133	0.7143 ±0.0162	0.7634 ±0.0151	0.7149 ±0.0086	0.6432 ±0.0055	0.7080 ±0.0039
SPaRCNet [18]	–	0.7487 ±0.0075	0.7097 ±0.0132	0.7624 ±0.0092	0.4756 ±0.1109	0.3147 ±0.1315	0.4108 ±0.1310
ContraWR [37]	–	0.7402 ±0.0126	0.7178 ±0.0156	0.7610 ±0.0137	0.4242 ±0.0541	0.2340 ±0.0554	0.2987 ±0.0288
CNN-Transformer [26]	–	0.7363 ±0.0087	0.7129 ±0.0121	0.7719 ±0.0153	0.6573 ±0.0141	0.5961 ±0.0105	0.6896 ±0.0065
FFCL [23]	–	0.7277 ±0.0182	0.7016 ±0.0292	0.7614 ±0.0197	0.4427 ±0.0702	0.2542 ±0.0654	0.2902 ±0.0485
ST-Transformer [28]	–	0.7381 ±0.0205	0.7013 ±0.0352	0.7681 ±0.0175	0.2559 ±0.0141	0.0503 ±0.0183	0.1428 ±0.0122
Foundation models							
BIOT [36]	mixed	0.7527 ±0.0121	0.7291 ±0.0230	0.7790 ±0.0146	0.6862 ±0.0041	0.6295 ±0.0113	0.7091 ±0.0147
LaBraM-Base [16]	∼2.5k	0.7633 ±0.0102	0.7231 ±0.0182	0.7810 ±0.0133	0.7277 ±0.0101	0.6813 ±0.0053	0.7454 ±0.0027
CBraMod [33]	∼9k	0.7865 ±0.0110	0.7442 ±0.0152	0.8011 ±0.0099	0.7269 ±0.0041	0.6685 ±0.0104	0.7395 ±0.0089
CSBrain [39]	∼9k	0.7925 ±0.0030	0.7406 ±0.0102	0.7990 ±0.0091	0.7345 ±0.0047	0.6818 ±0.0046	0.7506 ±0.0042
REVE [11]	∼60k	0.7819	0.7500	0.8005	–	–	–
Raw-waveform recon. (same arch.)	2,388	0.8062 ±0.0061	0.7519 ±0.0097	0.8100 ±0.0069	0.7551 ±0.0079	0.6935 ±0.0057	0.7631 ±0.0049
SpecBraM	2,388	**0.8177** ±0.0032	**0.7670** ±0.0070	**0.8198** ±0.0051	**0.7696** ±0.0012	**0.7151** ±0.0021	**0.7796** ±0.0024

Table 20. **Full fine-tuning beyond sleep** (balanced accuracy, mean±std over three pretraining seeds). Same samples and splits as the linear probe; encoder and a linear head on mean-pooled tokens are trained with layer-wise learning-rate decay, and the epoch is selected on validation balanced accuracy. Bold: best per column.

Model	MA	Mumtaz	FACED	PhysioNet-MI	TUAB	TUEV
SpecBraM	**0.740** ±0.109	0.928 ±0.026	**0.276** ±0.017	**0.564** ±0.027	**0.821** ±0.004	**0.650** ±0.079
Raw-waveform recon. (same arch.)	0.670 ±0.018	**0.936** ±0.006	0.273 ±0.013	0.531 ±0.005	0.792 ±0.006	0.566 ±0.025

Table 21. **Spectral estimator of the band-power target** (linear patch embedding; everything else as the main model). Strict linear probe on sleep staging, mean±std over three pretraining seeds unless stated; columns as Table 6. Bold: best per column.

	BA (100% labels)		BA (1% labels)		Δ BA (pts)	
Target	ISRUC	HMC	ISRUC	HMC	ISRUC	HMC
Band power (main: periodogram, five classical bands)	**0.7965** ±0.0059	0.7433 ±0.0019	0.765 ±0.006	0.670 ±0.008	–	–
multitaper estimate (3 DPSS tapers, $NW{=}2$)	0.7926 ±0.0017	**0.7499** ±0.0064	**0.770** ±0.002	**0.686** ±0.005	-0.4	+0.7

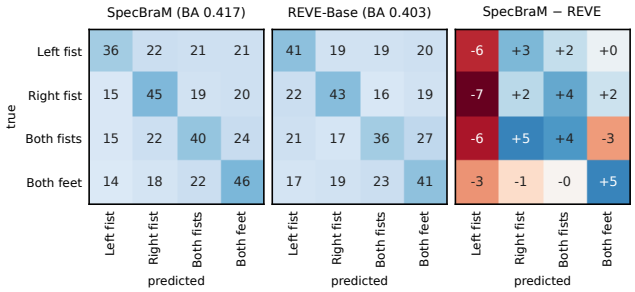


Figure 14. **Confusion matrices on PhysioNet-MI** (strict linear probe, row-normalized recall in %): SpecBraM's main checkpoint, REVE-Base and their difference (blue: SpecBraM higher).

Table 22. **Factors besides the target in the comparison with external models** (strict linear probe, balanced accuracy; our rows: mean±std over pretraining seeds). Per-channel readout: tokens averaged over time within each channel and concatenated; still a frozen encoder and a logistic regression. Our normalization for CBraMod/LaBraM: per-recording median/IQR scaling, rescaled to the dataset's typical amplitude.

	ISRUC		HMC	
Model	(a)	(b)	(a)	(b)
Readout: (a) mean over all tokens (main protocol), (b) per-channel means				
SpecBraM	0.7965 ±0.0059	0.7957 ±0.0033	0.7433 ±0.0019	0.7504 ±0.0019
Raw-waveform recon.	0.7701 ±0.0076	0.7734 ±0.0039	0.7153 ±0.0028	0.7220 ±0.0041
CBraMod	0.7278	0.7708	0.6190	0.6841
LaBraM-Base	0.7523	0.7644	0.6323	0.6950
CSBrain	0.7304	0.7654	0.6114	0.6517
REVE-Base	0.7928	0.7995	0.7040	0.7223
Input: (a) the model's own convention, (b) our per-recording normalization				
CBraMod	0.7278	0.7413	0.6190	0.6732
LaBraM-Base	0.7523	0.7326	0.6323	0.6524
Scale: (a) 51.3M parameters, (b) 4.2M parameters (CBraMod: 4.0M)				
Band power	0.7965 ±0.0059	0.7882 ±0.0053	0.7433 ±0.0019	0.7284 ±0.0040
Raw-waveform recon.	0.7701 ±0.0076	0.7537 ±0.0042	0.7153 ±0.0028	0.6738 ±0.0083

Table 23. **Handcrafted spectral features under the same strict linear probe** (balanced accuracy). Basic: log absolute and relative power in the five classical bands per channel. Rich: 1-Hz log PSD (0.5–45 Hz), relative power in six bands, Hjorth parameters, spectral entropy, 95% spectral edge, line length and kurtosis per channel. Features are computed on our input of each window (sleep: averaged over the three 10-s windows of an epoch, as for every encoder). Bold: best per column.

	Sleep, 100%		Sleep, 1%		Beyond sleep					
Features	ISRUC	HMC	ISRUC	HMC	MA	Mumtaz	FACED	PhysioNet-MI	TUAB	TUEV
SpecBraM	**0.7965** ±0.0059	**0.7433** ±0.0019	**0.765** ±0.006	0.670 ±0.008	**0.747** ±0.030	**0.942** ±0.007	0.216 ±0.013	0.397 ±0.018	**0.805** ±0.007	**0.561** ±0.025
Raw-waveform recon. (same arch.)	0.7701 ±0.0076	0.7153 ±0.0028	0.721 ±0.017	0.617 ±0.017	0.685 ±0.051	0.924 ±0.013	**0.220** ±0.010	0.395 ±0.013	0.804 ±0.002	0.533 ±0.027
Handcrafted: 5-band log/relative power	0.7418	0.6825	0.707	0.624	0.663	0.926	0.201	0.391	0.735	0.498
Handcrafted: rich spectral set	0.7636	0.7227	0.747	**0.674**	0.712	0.906	0.217	**0.410**	0.752	0.469